

Biochemistry II

Carbohydrate and lipid metabolism, nitrogen pathways and integrated clinical reasoning

Updated 8 September 2026

Local lecture evidence covers glycolysis, fermentation, other sugars, glycogen, gluconeogenesis, TCA and dietary lipid digestion/absorption with chylomicron disorders. Important related mechanisms are expanded using the local Ferrier 6th edition. Nitrogen metabolism, complex lipids, nucleotide metabolism, vitamins and extended integration are explicitly book extensions because corresponding Term 2 lectures are not present in the supplied course folder. The local file is 6th edition, although the lipid lecturer names 7th edition. This is a detailed review of available material, not certification of the university exam blueprint.

How to use this book

Lecture-grounded material and book extensions are distinguished. Green/blue panels consolidate the key ideas; amber panels flag missing evidence; orange panels correct traps. This is a detailed review of the available sources, not an official exam blueprint. Original labeled figures remain important. Numerical laboratory ranges and procedures depend on the course method and local SOP.

Original lecture transcripts are preserved as PDF attachments and in the editable source folder when available. Automated transcripts can contain errors; study the checked synthesis first. Personal study only: original book/lecture illustrations retain their owners' rights.

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Reading map and source confidence

SOURCE BASIS: lecture + book

Use the lecture-supported carbohydrate and dietary-lipid sections first. Book extensions supply the mechanistic background and clinical comparisons needed to reason through difficult questions, but their inclusion here does not establish that every extension is on the exam. The transcripts remain indexed as source material. Their repetitions and verbal slips have been condensed into checked explanations.

Topic	Local evidence	Local Lippincott PDF reading
Glycolysis, other sugars, glycogen, gluconeogenesis, TCA	Seven handwritten pages, reconstructed Part 2 and three carbohydrate recordings	Chs. 8-12, pp. 176-276
Bioenergetics and pentose phosphate pathway	Mentioned or required to interpret carbohydrate lectures; expanded from book	Ch. 6, pp. 135-160; Ch. 13, pp. 277-297
Dietary lipids, absorption and chylomicrons	Three parts of Esmaeili lipid lecture 1	Ch. 15, pp. 326-339; relevant Ch. 18, pp. 420-432
Fatty acids, ketones, complex lipids and cholesterol	Book extension beyond the available first lipid lecture	Chs. 16-18, pp. 340-447
Nitrogen, amino acids and specialized products	Book extension; no corresponding course lecture found	Chs. 19-21, pp. 448-531
Nucleotides and integrated metabolism	Book extension; relevance inferred from requested Term 2 metabolism scope	Chs. 22-26, pp. 532-650
Nutrition and vitamin mechanisms	Targeted book extension, not the full nutrition syllabus	Chs. 27-28, pp. 651-720

Corrections to remember

Glycolysis itself does not consume oxygen. Each pyruvate has three carbons. Fructose 2,6-bisphosphate activates PFK-1, whereas fructose 1,6-bisphosphate feeds forward to pyruvate kinase. The urea cycle uses three ATP molecules but four high-energy phosphate bonds. Acetyl-CoA from even-chain fatty acids cannot provide net glucose in humans.

Sources for this section

Biochemistry Metabolism (Part 1) Notes - Hawraa.pdf - PDF pp. 1-7; all handwritten pages visually inspected

Biochemistry metabolism notes (pt 2) slides 27-80 .pdf - PDF pp. 1-6

IMG_4406.transcript.txt - entire transcript

IMG_4407.transcript.txt - entire transcript

Biochemistry Metabolism Recorded Lecture - 1h04m20s.transcript.txt - entire transcript

Lipid metabolism Esmaeili 1.1.transcript.txt - entire transcript

Lipid metabolism Esmaeili 1.2.transcript.txt - entire transcript

Lipid metabolism1 Esmaeili 1.3.transcript.txt - entire transcript

Compartmentation, energy and pathway regulation

SOURCE BASIS: lecture + book

Metabolism is an interconnected network. Catabolic pathways release usable energy and reducing equivalents while breaking molecules into smaller products. Anabolic pathways use ATP and reducing power to build molecules. A pathway may be amphibolic: the TCA cycle oxidizes acetyl units but also provides intermediates for biosynthesis. Flux depends on substrate supply, enzyme activity, compartment access and demand for products. A reaction can be thermodynamically favorable yet slow if its activation barrier is high; enzymes lower the barrier without changing the reaction equilibrium.

Location	Major processes	Consequence
Cytosol	Glycolysis, pentose phosphate pathway, glycogen metabolism, fatty-acid synthesis	NADPH made in the cytosol can support reductive synthesis and antioxidant defense.
Mitochondrial matrix	PDH, most TCA enzymes, beta-oxidation, ketogenesis, pyruvate carboxylase, first two urea reactions	Mature RBCs cannot perform these pathways.
Inner mitochondrial membrane	Respiratory chain, ATP synthase, succinate dehydrogenase, specific carriers	An intact membrane and proton gradient couple oxidation to ATP synthesis.
ER	Glucose-6-phosphatase system, cholesterol regulation, fatty-acid elongation/desaturation, lipid assembly	Metabolite transport into or across this compartment can be a separate control point.
Lysosomes/peroxisomes	Macromolecule degradation / very-long-chain and branched-chain fatty-acid processing	A compartment defect can block a pathway even when the catalytic enzyme is otherwise present.

- High ATP and NADH generally signal abundant energy; AMP/ADP and NAD⁺ favor fuel oxidation. These are tendencies applied to particular enzymes, not universal on/off rules.
- NADH mainly conveys electrons toward energy production. NADPH mainly supports reductive biosynthesis, glutathione recycling, P450 reactions and the phagocyte oxidative burst. They are distinct coenzyme pools.
- Allosteric regulation can act rapidly. Reversible phosphorylation couples hormone signals to enzyme activity. Changes in transcription or protein degradation alter enzyme amount more slowly.
- Insulin binds a receptor tyrosine kinase. Hepatic glucagon signaling uses a G-protein-coupled receptor, adenylyl cyclase, cAMP and protein kinase A. Phosphorylation activates some enzymes and inhibits others.

Sources for this section

Biochemistry Metabolism (Part 1) Notes - Hawraa.pdf - PDF pp. 1-7; all handwritten pages visually inspected

Biochemistry metabolism notes (pt 2) slides 27-80 .pdf - PDF pp. 1-6

IMG_4406.transcript.txt - entire transcript

IMG_4407.transcript.txt - entire transcript

Biochemistry Metabolism Recorded Lecture - 1h04m20s.transcript.txt - entire transcript

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - Ch. 6 pp. 135-158; Ch. 8 pp. 176-184; Ch. 24 pp. 584-608

Glucose transport, trapping and tissue priorities

SOURCE BASIS: lecture + book

Transport and phosphorylation are separate steps. Facilitated GLUT transport moves glucose down its concentration gradient. Phosphorylation traps the sugar as glucose-6-phosphate and keeps free intracellular glucose low. In the intestine and kidney, sodium-coupled glucose transport can move glucose against its gradient using the sodium gradient maintained by Na^+/K^+ -ATPase.

Protein	Main teaching sites	Important distinction
GLUT1	RBCs, blood-brain barrier and widespread basal uptake	Insulin-independent basal transport.
GLUT3	Neurons	High-affinity uptake supports neuronal glucose use.
GLUT2	Liver, basolateral intestine/kidney; classic book pancreatic model	High capacity, bidirectional movement as gradients change. Hepatic uptake is not GLUT4-mediated.
GLUT4	Skeletal/cardiac muscle and adipose tissue	Insulin recruits transporters to the surface. Muscle contraction also promotes uptake.
GLUT5	Intestinal apical fructose transport	Fructose transport is not glucose transport.

Property	Hexokinase	Glucokinase
Affinity/capacity	Low K_m , lower V_{max} : functions at low glucose	Higher K_m and V_{max} : handles a carbohydrate load
Product inhibition	Inhibited by glucose-6-phosphate	Not directly inhibited by glucose-6-phosphate
Tissue purpose	Protects basal glucose use in many tissues	Hepatic glucose handling and pancreatic glucose sensing
Regulation	Local product/energy demand	Hepatic expression increased by insulin; glucokinase regulatory protein responds to sugar phosphates

Tissue use is the exam anchor

RBCs always rely on glycolysis for ATP. The brain ordinarily depends heavily on glucose but adapts to use ketones during prolonged fasting; this is not an instantaneous guarantee of protection in acute hypoglycemia. Liver maintains blood glucose. Muscle retains its glycogen for its own work.

Hepatic GKR binds and sequesters glucokinase in the nucleus. Fructose-6-phosphate favors formation of the inactive glucokinase-GKR complex, whereas fructose-1-phosphate opposes it. Increased glucose also favors release and cytosolic activity. This regulation is not direct feedback inhibition of glucokinase by glucose-6-phosphate. Glucokinase inactivation can cause MODY2 by shifting glucose sensing and insulin secretion thresholds.

Sources for this section

Biochemistry Metabolism (Part 1) Notes - Hawraa.pdf - PDF pp. 1-7; all handwritten pages visually inspected

Biochemistry metabolism notes (pt 2) slides 27-80 .pdf - PDF pp. 1-6

IMG_4406.transcript.txt - entire transcript

IMG_4407.transcript.txt - entire transcript

Biochemistry Metabolism Recorded Lecture - 1h04m20s.transcript.txt - entire transcript

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - Ch. 8 pp. 185-190; summary pp. 202-203

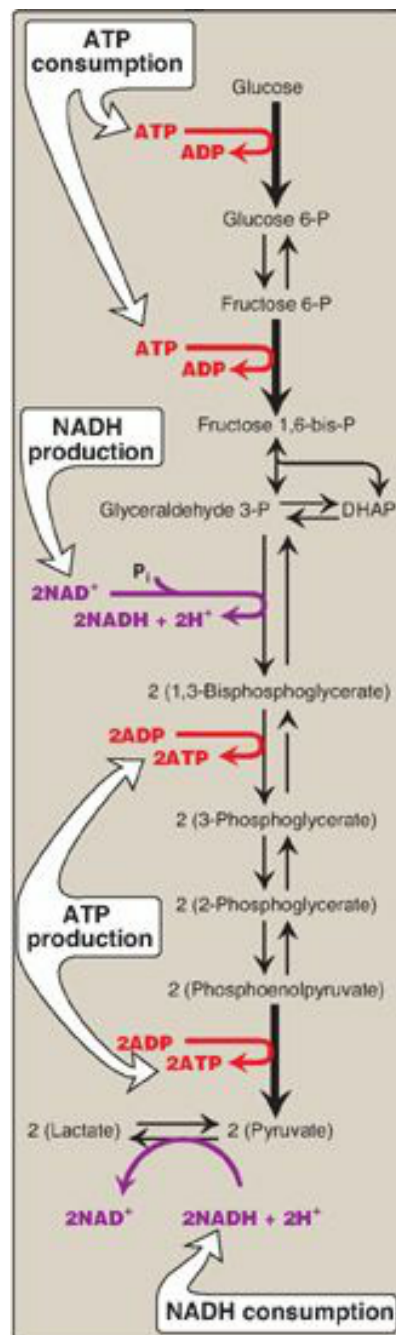
Glycolysis: every step, carbon count and energy event

SOURCE BASIS: lecture + book

One six-carbon glucose forms two three-carbon pyruvates in the cytosol. The first phase spends two ATP. Cleavage and isomerization create two glyceraldehyde-3-phosphates, so every subsequent reaction occurs twice per starting glucose. The payoff phase produces four ATP and two NADH, giving a net two ATP. Molecular oxygen is not a reactant in any glycolytic step. Under aerobic conditions the reducing equivalents can be reoxidized through mitochondrial shuttles and the respiratory chain; under oxygen limitation or in RBCs, lactate formation regenerates cytosolic NAD⁺.

Step and enzyme	Conversion	Energy or key property
1 Hexokinase/glucokinase	Glucose to glucose-6-phosphate	Consumes ATP; physiologically irreversible.
2 Phosphoglucose isomerase	Glucose-6-phosphate to fructose-6-phosphate	Aldose/ketose interconversion.
3 PFK-1	Fructose-6-phosphate to fructose-1,6-bisphosphate	Consumes ATP; committed, regulated, irreversible step.
4 Aldolase	Fructose-1,6-bisphosphate to GAP plus DHAP	Six carbons become two three-carbon compounds.
5 Triose phosphate isomerase	DHAP to GAP	Two GAP molecules proceed through payoff.
6 GAP dehydrogenase	GAP plus Pi to 1,3-bisphosphoglycerate	NAD ⁺ becomes NADH; no ATP supplies this added phosphate.
7 Phosphoglycerate kinase	1,3-BPG to 3-phosphoglycerate	Substrate-level phosphorylation produces ATP.
8 Phosphoglycerate mutase	3-phosphoglycerate to 2-phosphoglycerate	Moves the existing phosphate within the molecule.

Step and enzyme	Conversion	Energy or key property
9 Enolase	2-phosphoglycerate to phosphoenolpyruvate	Dehydration creates high transfer-potential PEP.
10 Pyruvate kinase	PEP to pyruvate	Substrate-level phosphorylation; irreversible.



Source pathway: trace the two ATP investments, two doubled ATP-producing steps and regeneration of NAD⁺ during lactate formation. The diagram is a compact source image; use the step table for enzyme names.

Identify: Glucose-6-phosphate; Fructose-1,6-bisphosphate; DHAP/GAP; 1,3-BPG; PEP; Pyruvate; Lactate

Figure source: Lippincott Biochemistry 6e, PDF p. 198, extracted source diagram

Check yourself: If GAP dehydrogenase lacks NAD⁺, can abundant glucose and ADP keep glycolysis running?

No. Oxidation of GAP cannot proceed, so the downstream ATP-producing steps are starved of intermediates. Lactate dehydrogenase normally recycles NADH to NAD⁺ under anaerobic conditions. ATP availability alone does not replace the required electron acceptor.

Sources for this section

Biochemistry Metabolism (Part 1) Notes - Hawraa.pdf - PDF pp. 1-7; all handwritten pages visually inspected

Biochemistry metabolism notes (pt 2) slides 27-80 .pdf - PDF pp. 1-6

IMG_4406.transcript.txt - entire transcript

IMG_4407.transcript.txt - entire transcript

Biochemistry Metabolism Recorded Lecture - 1h04m20s.transcript.txt - entire transcript

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - Ch. 8 pp. 188-204

PFK-1, pyruvate kinase and the lactate shunt

SOURCE BASIS: lecture + book

Control	Activation	Inhibition and implication
PFK-1	AMP and fructose-2,6-bisphosphate	ATP and citrate suppress flux when energy and biosynthetic carbon are abundant.
Hepatic PFK-2/FBPase-2	Insulin favors dephosphorylation, raising fructose-2,6-bisphosphate	Glucagon/PKA phosphorylation lowers fructose-2,6-bisphosphate in the liver, opposing glycolysis.
Pyruvate kinase	Fructose-1,6-bisphosphate feed-forward activation	ATP and alanine inhibit; hepatic glucagon/PKA phosphorylation inhibits.
Hexokinase	Substrate availability	Glucose-6-phosphate feedback inhibition.

Lactate dehydrogenase catalyzes pyruvate + NADH + H⁺ to lactate + NAD⁺. This reversible reaction conserves the carbon skeleton while regenerating NAD⁺. It produces no additional ATP. In the Cori cycle, lactate travels to liver for gluconeogenesis, transferring part of the energetic burden to liver. Alcoholic fermentation makes acetaldehyde and then ethanol in microorganisms. Humans do not use pyruvate decarboxylase to ferment glucose to ethanol.

In RBCs, diversion of 1,3-BPG through the 2,3-BPG shunt changes hemoglobin oxygen affinity while bypassing an ATP-producing reaction. Pyruvate kinase deficiency reduces RBC ATP and can cause hemolysis. This differs from G6PD deficiency, where the primary failure is NADPH-dependent antioxidant protection, not a direct ATP-producing step.

Correcting the lecture transcription

The early transcript and handwritten p.6 describe PFK inhibition by fructose-1,6-bisphosphate. The later lecture correctly discusses ATP and citrate inhibition. The standard distinction is PFK-1 activation by fructose-2,6-bisphosphate versus pyruvate-kinase activation by fructose-1,6-bisphosphate. Do not memorize the early error.

Sources for this section

Biochemistry Metabolism (Part 1) Notes - Hawraa.pdf - PDF pp. 1-7; all handwritten pages visually inspected

Biochemistry metabolism notes (pt 2) slides 27-80 .pdf - PDF pp. 1-6

IMG_4406.transcript.txt - entire transcript

IMG_4407.transcript.txt - entire transcript

Biochemistry Metabolism Recorded Lecture - 1h04m20s.transcript.txt - entire transcript

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - Ch. 8 pp. 190-203

PDH and the TCA cycle: enzymes, cofactors and regulation

SOURCE BASIS: lecture + book

Pyruvate enters mitochondria and undergoes irreversible oxidative decarboxylation by the pyruvate dehydrogenase complex: $\text{pyruvate} + \text{CoA} + \text{NAD}^+ \rightarrow \text{acetyl-CoA} + \text{CO}_2 + \text{NADH}$. The two-carbon acetyl group enters the TCA cycle by condensing with four-carbon oxaloacetate. PDH is a bridge reaction, not one of the eight cycle reactions. Its cofactors are TPP (B1), lipoate, CoA (B5), FAD (B2) and NAD^+ (B3). The alpha-ketoglutarate and branched-chain ketoacid dehydrogenase complexes use the same five-cofactor pattern.

TCA enzyme	Reaction	Output/control
Citrate synthase	Oxaloacetate + acetyl-CoA to citrate	Condensation; citrate can inhibit.
Aconitase	Citrate through cis-aconitate to isocitrate	Isomerization.
Isocitrate dehydrogenase	Isocitrate to alpha-ketoglutarate	$\text{CO}_2 + \text{NADH}$; ADP/ Ca^{2+} stimulate, ATP/NADH inhibit.
Alpha-ketoglutarate dehydrogenase	Alpha-ketoglutarate to succinyl-CoA	$\text{CO}_2 + \text{NADH}$; Ca^{2+} stimulates, NADH/succinyl-CoA inhibit.
Succinyl-CoA synthetase	Succinyl-CoA to succinate	GTP or ATP depending on isoform; substrate-level phosphorylation.
Succinate dehydrogenase	Succinate to fumarate	FAD-linked oxidation; also respiratory complex II in inner membrane.
Fumarase	Fumarate to malate	Hydration.
Malate dehydrogenase	Malate to oxaloacetate	NADH; oxaloacetate consumption helps pull the reaction forward.

- Per acetyl-CoA, the cycle yields three NADH, one FADH_2 , one GTP and two CO_2 . The two newly entering acetyl carbons are not necessarily the two carbons lost as CO_2 on the first turn.
- PDH kinase phosphorylates and inhibits PDH. High ATP, NADH and acetyl-CoA favor this inhibition. PDH phosphatase removes the phosphate; calcium stimulates activity in working muscle.
- The cycle does not directly consume molecular oxygen, but depends on the respiratory chain to regenerate oxidized cofactors. Respiratory failure increases NADH and slows TCA flux.
- Pyruvate carboxylase replenishes oxaloacetate using biotin, ATP and bicarbonate. Acetyl-CoA activates it, coordinating incoming acetyl units with availability of the accepting four-carbon pool.

- Citrate can leave for fatty-acid synthesis; alpha-ketoglutarate and oxaloacetate support amino-acid synthesis; succinyl-CoA supports heme synthesis. Withdrawal must be balanced by anaplerosis.

Why PDH or thiamine deficiency raises lactate

When pyruvate cannot be efficiently oxidized, more is reduced to lactate. ATP supply to highly oxidative tissues falls. Arsenic can interfere with lipoate-dependent enzyme complexes. These are mechanism comparisons, not diagnostic conclusions from lactate alone.

PDH component	Role	Cofactor link
E1, pyruvate dehydrogenase	Decarboxylates pyruvate and transfers the hydroxyethyl equivalent	TPP
E2, dihydrolipoyl transacetylase	Oxidizes/transfers the acetyl group to CoA using a swinging lipoyl arm	Lipoate and CoA
E3, dihydrolipoyl dehydrogenase	Reoxidizes reduced lipoate and transfers reducing equivalents onward	FAD then NAD+

Arsenite inhibits lipoate-dependent complexes, whereas arsenate can substitute for phosphate in the GAP-dehydrogenase reaction and bypass capture of energy at phosphoglycerate kinase. Distinguish these chemical forms and sites rather than saying every arsenic question has the same mechanism.

Sources for this section

Biochemistry Metabolism (Part 1) Notes - Hawraa.pdf - PDF pp. 1-7; all handwritten pages visually inspected

Biochemistry metabolism notes (pt 2) slides 27-80 .pdf - PDF pp. 1-6

IMG_4406.transcript.txt - entire transcript

IMG_4407.transcript.txt - entire transcript

Biochemistry Metabolism Recorded Lecture - 1h04m20s.transcript.txt - entire transcript

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - Ch. 9 pp. 209-223

Respiratory chain and transparent ATP accounting

SOURCE BASIS: lecture + book

Electrons from NADH enter complex I. Electrons from succinate oxidation enter through complex II. Coenzyme Q transfers electrons to III, cytochrome c to IV, and IV reduces oxygen to water. Complexes I, III and IV pump protons; II does not. ATP synthase uses proton return to the matrix. The chain and phosphorylation normally run together because the proton gradient restrains further pumping when it is not dissipated.

Perturbation	Electron flow / oxygen use	ATP and gradient
Respiratory-chain inhibitor	Electron flow and oxygen consumption fall downstream of the block	ATP production falls.
ATP synthase inhibitor	Backpressure from the gradient slows respiratory electron flow	ATP falls while the proton gradient initially rises.

Perturbation	Electron flow / oxygen use	ATP and gradient
Uncoupler or UCP1	Protons return without ATP synthase; respiration can increase when substrate is available	Less ATP per oxygen consumed; energy appears as heat.

Source per glucose	Products	ATP equivalents at NADH 2.5, FADH2 1.5
Glycolysis	2 net ATP + 2 cytosolic NADH	2 + 3 or 5, depending on shuttle
Two PDH reactions	2 NADH	5
Two TCA turns	6 NADH + 2 FADH2 + 2 GTP	15 + 3 + 2 = 20
Complete oxidation	Sum of all stages	30 or 32 ATP, theoretical bookkeeping

Edition convention matters

The inspected 6th edition uses older values of three ATP per NADH and two per FADH2 in some chapters. The lecture uses 30-32 ATP per glucose. This review uses 2.5/1.5 for modern worked calculations and states when an old-book answer differs. Actual cellular yields vary with transport costs, proton leak and physiological conditions.

Agent/site	Direct effect	Predict the downstream response
Rotenone or amytal: complex I	Blocks NADH-derived electron transfer toward CoQ	Succinate-derived entry through complex II can bypass the I block in an appropriate experimental system.
Antimycin A: complex III	Blocks electron transfer through cytochrome bc1	NADH- and succinate-derived paths converge before this block, so simply changing between them does not bypass III.
Cyanide, carbon monoxide, hydrogen sulfide or azide: complex IV	Interferes with terminal reduction of oxygen	Upstream carriers tend to become reduced; oxidative ATP supply falls.
Oligomycin: ATP-synthase Fo channel	Blocks proton return through synthase	ATP synthesis falls; the accumulating gradient slows respiration.
2,4-dinitrophenol or UCP1	Dissipates gradient without ATP capture	Electron transfer can accelerate but ATP efficiency falls and heat rises.

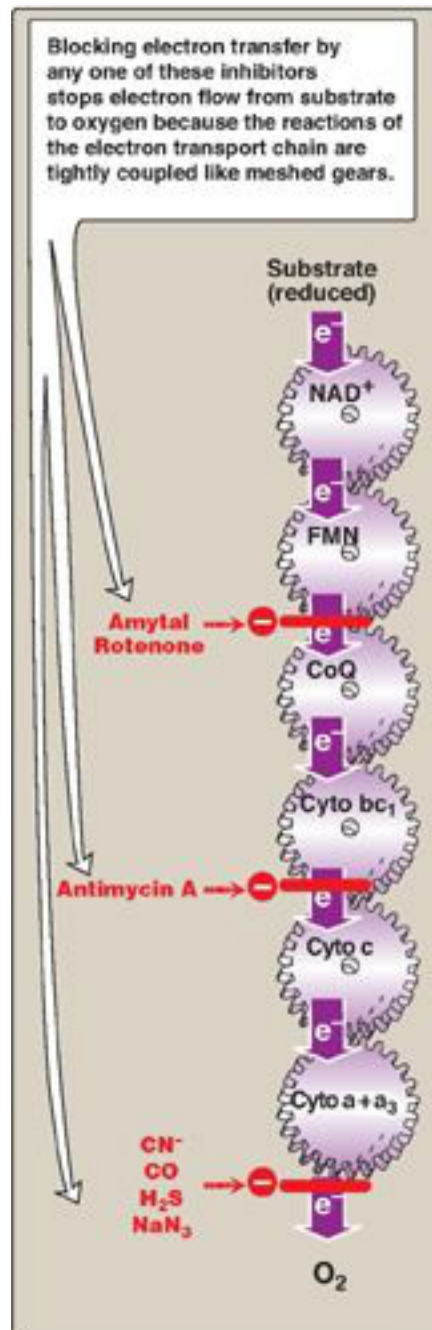


Figure 6.10 from the local book: site-specific inhibitor map. Use it to locate the block, then predict oxidation states on either side; do not merely memorize a poison name.

Identify: Complex I; Coenzyme Q; Complex III; Cytochrome c; Complex IV

Figure source: Lippincott Biochemistry 6e, PDF p. 149, extracted source diagram

Sources for this section

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - Ch. 6 pp. 135-158; Ch. 9 p.220

Biochemistry Metabolism Recorded Lecture - 1h04m20s.transcript.txt - final TCA explanation and 30-32 ATP discussion

Fructose, galactose, mannose and the polyol pathway

SOURCE BASIS: lecture + book

Hepatic fructokinase consumes ATP to form fructose-1-phosphate. Aldolase B cleaves this to DHAP and glyceraldehyde, and triose kinase phosphorylates glyceraldehyde to GAP. Entry below PFK-1 bypasses that major glycolytic control point. Hexokinase can form fructose-6-phosphate in some tissues but generally prefers glucose. Fructokinase deficiency causes relatively benign essential fructosuria. Aldolase B deficiency traps phosphate as fructose-1-phosphate, reducing ATP and impairing glycogenolysis/gluconeogenesis: hereditary fructose intolerance becomes apparent after fructose, sucrose or sorbitol exposure.

Galactose is the C4 epimer of glucose. Galactokinase makes galactose-1-phosphate. GALT transfers a uridyl group from UDP-glucose: galactose-1-phosphate + UDP-glucose produces UDP-galactose + glucose-1-phosphate. UDP-galactose 4-epimerase regenerates UDP-glucose. Phosphoglucomutase converts glucose-1-phosphate to glucose-6-phosphate. The reconstructed notes abbreviate this pathway so heavily that the UDP-glucose partner and released glucose-1-phosphate can be missed.

Defect/pathway	Biochemical consequence	Distinguishing pattern
Galactokinase deficiency	Galactose diverted to galactitol	Cataracts with a milder systemic picture.
Classic GALT deficiency	Galactose-1-phosphate accumulation and galactitol production	Illness after milk feeding, liver dysfunction and cataracts; different from lactose intolerance.
Lactase deficiency	Lactose remains in intestinal lumen	Osmotic diarrhea and bacterial fermentation, without a primary intracellular galactose-enzyme defect.
Aldose reductase	Glucose reduced to sorbitol using NADPH	Sorbitol accumulates in tissues with limited sorbitol dehydrogenase during hyperglycemia.
Sorbitol dehydrogenase	Sorbitol to fructose, making NADH	Prominent in tissues such as liver and seminal vesicles.
Mannose	Hexokinase makes mannose-6-P, then isomerization to fructose-6-P	Connects glycoprotein carbohydrate metabolism to glycolysis.

Check yourself: Why is the intact GALT reaction more informative than writing galactose-1-P to UDP-galactose?

The intact equation shows both UDP-glucose consumption and glucose-1-phosphate release. It explains the transferase name, how glucose carbon enters mainstream metabolism, and why epimerase is needed to replenish the UDP-glucose pool.

Lactose synthesis occurs in the Golgi of the lactating mammary gland. Lactose synthase combines a broadly expressed beta-galactosyltransferase with alpha-lactalbumin, whose expression is stimulated by prolactin. Alpha-lactalbumin changes the transferase substrate preference so it transfers galactose from UDP-galactose to glucose, producing lactose plus UDP. Without this modifying protein, the transferase preferentially forms N-acetyllactosamine using N-acetylglucosamine. This is a synthesis mechanism, separate from intestinal lactase digestion.

Sources for this section

Biochemistry metabolism notes (pt 2) slides 27-80 .pdf - PDF pp. 1-2

Biochemistry Metabolism Recorded Lecture - 1h04m20s.transcript.txt - opening fructose/galactose section

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - Ch. 12 pp. 262-276

Glycogen structure, synthesis, degradation and control

SOURCE BASIS: lecture + book

Glycogen is a branched glucose polymer with alpha-1,4 chains and alpha-1,6 branch points. Its numerous nonreducing ends permit rapid simultaneous synthesis and breakdown. Glycogenin creates the primer. UDP-glucose supplies activated glucose to glycogen synthase, which makes alpha-1,4 links. Branching enzyme transfers a chain segment to form an alpha-1,6 branch. Adding a free glucose to glycogen consumes an ATP for glucose phosphorylation and a UTP equivalent for UDP-glucose formation.

Glycogen phosphorylase uses inorganic phosphate, not water, to remove alpha-1,4-linked residues as glucose-1-phosphate. It requires PLP and stops four residues before a branch. The debranching transferase moves three residues, then alpha-1,6-glucosidase releases the remaining branch-point glucose as free glucose. Most residues therefore emerge already phosphorylated; the small branch-point fraction is free glucose. In muscle, glycolysis of a glycogen-derived glucose-6-phosphate can net three ATP to lactate because the hexokinase ATP cost was avoided.

Situation	Liver	Skeletal muscle
Purpose of glycogen	Maintain blood glucose	Provide fuel for contraction
Glucose-6-phosphatase	Present: can release free glucose	Absent: G6P retained for local metabolism
Glucagon response	Activates cAMP/PKA glycogenolysis	No direct physiological glucagon response
Epinephrine	Promotes glycogen mobilization	Promotes glycogen mobilization
Calcium / AMP	Calcium can activate phosphorylase kinase; glucose signals abundance	Contraction-derived Ca ²⁺ activates phosphorylase kinase; AMP activates phosphorylase

Reciprocal phosphorylation

Phosphorylation activates phosphorylase kinase and glycogen phosphorylase while inhibiting glycogen synthase. Insulin-promoted dephosphorylation reverses the storage/mobilization balance. Always name the enzyme before deciding whether phosphorylation activates it.

Storage disease	Defect	Mechanistic clinical contrast
Von Gierke, type I	Glucose-6-phosphatase system	Severe fasting hypoglycemia because final glucose release fails in both glycogenolysis and gluconeogenesis.

Storage disease	Defect	Mechanistic clinical contrast
Pompe, type II	Lysosomal acid alpha-glucosidase	Lysosomal glycogen accumulation, cardiomyopathy/myopathy; different compartment from cytosolic degradation.
Cori, type III	Debranching enzyme	Limit-dextrin accumulation and incomplete glycogen degradation.
Andersen, type IV	Branching enzyme	Abnormally poorly branched glycogen, especially damaging to liver.
McArdle, type V	Muscle glycogen phosphorylase	Exercise intolerance, cramps and possible myoglobinuria; liver glucose release preserved.
Hers, type VI	Liver glycogen phosphorylase	Hepatic glycogenolysis impaired; usually milder fasting phenotype than type I.

Calcium activates phosphorylase kinase by binding its calmodulin subunit, linking contraction to fuel release even before a large change in ATP. Muscle glycogen phosphorylase is allosterically activated by AMP and inhibited by ATP and glucose-6-phosphate. Hepatic phosphorylase responds to glucose availability. In synthesis, UDP-glucose pyrophosphorylase activates glucose-1-phosphate using UTP; pyrophosphate hydrolysis favors the reaction. Glycogenin autoglucosylates a tyrosine hydroxyl to start the primer.

Sources for this section

Biochemistry metabolism notes (pt 2) slides 27-80 .pdf - PDF p.3

Biochemistry Metabolism Recorded Lecture - 1h04m20s.transcript.txt - glycogen and GLUT2/GLUT4 discussion

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - Ch.11 pp.240-261

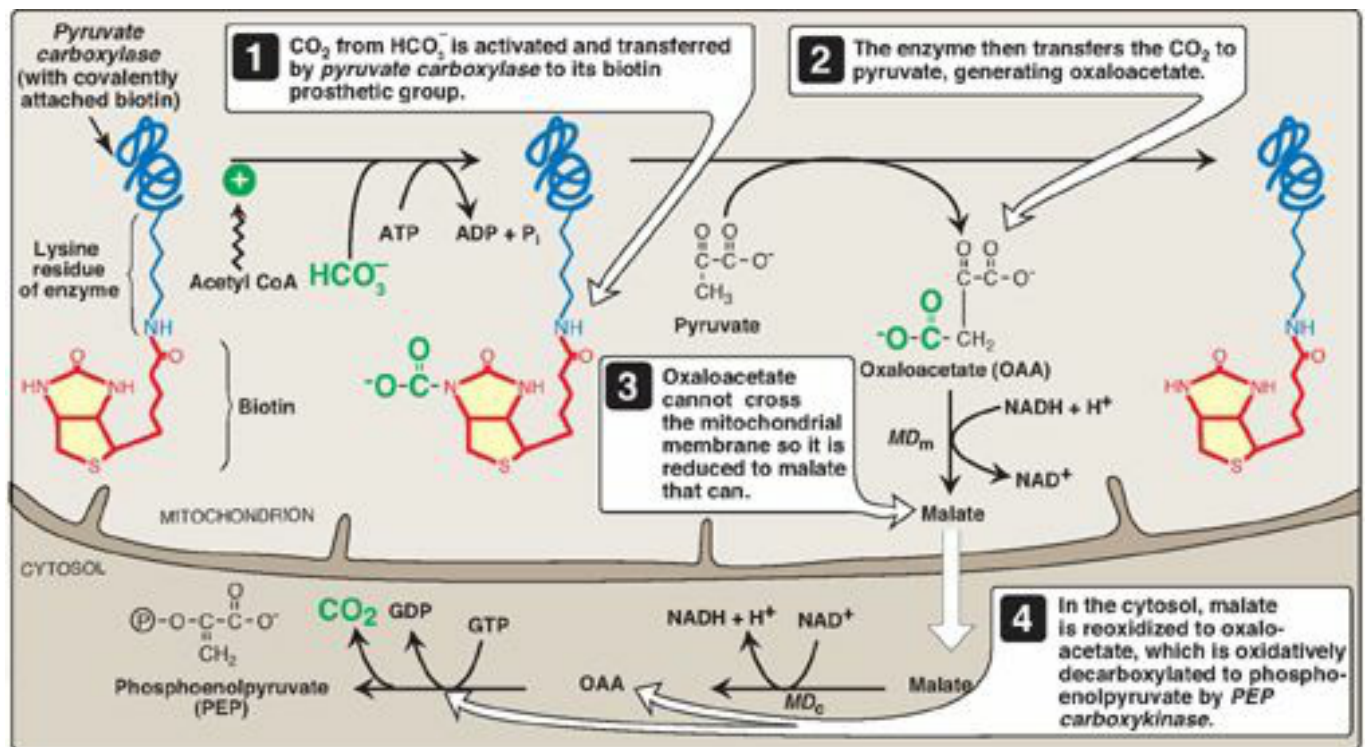
Gluconeogenesis, redox transfer and substrate limits

SOURCE BASIS: lecture + book

Liver is the main fasting glucose producer. Kidney cortex becomes increasingly important in prolonged fasting. Gluconeogenesis uses seven reversible glycolytic reactions plus bypasses for the three irreversible steps. Major substrates are lactate, glycerol and glucogenic amino-acid carbon skeletons. The glycerol backbone of TAG can yield glucose, but acetyl-CoA from its even-chain fatty acids cannot produce net glucose. Oxaloacetate is regenerated rather than created net when acetyl-CoA merely turns the TCA cycle.

Bypass	Enzyme/location	Requirement
Pyruvate to oxaloacetate	Pyruvate carboxylase, mitochondrial matrix	ATP, biotin and bicarbonate; acetyl-CoA activates.
Oxaloacetate to PEP	PEP carboxykinase; cytosolic and mitochondrial isoforms	GTP; releases the previously added CO ₂ .

Bypass	Enzyme/location	Requirement
Fructose-1,6-bisP to fructose-6-P	Fructose-1,6-bisphosphatase	Hydrolysis, not ATP generation. Inhibited by AMP and fructose-2,6-bisP.
Glucose-6-P to glucose	Glucose-6-phosphatase system in ER	Permits liver/kidney glucose export; absent from muscle.



Source diagram of the pyruvate-carboxylase and cytosolic-PEPCK route: track carbon, biotin and the malate shuttle separately. Other substrate-dependent routes may use mitochondrial PEPCK.

Identify: Pyruvate carboxylase; Biotin; Oxaloacetate; Malate; Cytosolic NADH; PEPCK

Figure source: Lippincott Biochemistry 6e, PDF p. 229, extracted source diagram

Oxaloacetate cannot directly cross the inner mitochondrial membrane. In the route emphasized by the lecture, reduction to malate allows export; cytosolic reoxidation restores oxaloacetate and provides NADH needed later in glucose synthesis. Lactate oxidation already supplies cytosolic NADH, so route selection can differ according to precursor. Glycerol enters via glycerol-3-phosphate then DHAP, downstream of the pyruvate bypass. From two pyruvates to one glucose, the requirement is four ATP, two GTP and two NADH.

Alcohol, fasting and hypoglycemia

Ethanol oxidation raises hepatic NADH/ NAD^+ . This favors pyruvate-to-lactate and oxaloacetate-to-malate conversion, reducing available gluconeogenic intermediates. The risk of hypoglycemia is especially important when glycogen stores are depleted. Glycerol is not restricted to glycolysis during starvation: the hormonal setting favors its contribution to glucose synthesis.

Sources for this section

Biochemistry metabolism notes (pt 2) slides 27-80 .pdf - PDF pp.4-5

Biochemistry Metabolism Recorded Lecture - 1h04m20s.transcript.txt - gluconeogenesis, malate and energy discussion

Pentose phosphate pathway and antioxidant defense

SOURCE BASIS: lecture + book

The cytosolic pentose phosphate pathway serves two demands: NADPH and ribose-5-phosphate. Its oxidative phase converts glucose-6-phosphate to ribulose-5-phosphate, releasing CO₂ and producing two NADPH. G6PD controls entry and responds to the NADP⁺/NADPH balance. The nonoxidative phase reversibly exchanges sugar carbons through transketolase and transaldolase, linking pentoses to fructose-6-phosphate and GAP. Transketolase requires TPP. No ATP is directly made by the pathway.

Cellular need	Pathway strategy	Reasoning
Ribose exceeds NADPH demand	Reverse nonoxidative reactions from glycolytic intermediates	Build ribose without obligatory oxidative NADPH production.
NADPH and ribose both needed	Oxidative phase followed by pentose isomerization	A biosynthetic cell can use both products.
NADPH exceeds ribose demand	Recycle pentoses into glycolytic intermediates and regenerate hexose phosphate	Keep extracting reducing power without accumulating unused ribose.

Glutathione peroxidase uses reduced glutathione to remove peroxides. Glutathione reductase uses NADPH to replenish reduced glutathione. RBCs have no mitochondria and depend strongly on the PPP for this defense. G6PD deficiency predisposes to oxidative hemolysis, with damaged hemoglobin forming Heinz bodies and splenic removal producing bite cells. Oxidant exposure, fava beans or infection can trigger episodes.

The same reductant can protect cells or help kill microbes: phagocyte NADPH oxidase forms superoxide, superoxide dismutase forms hydrogen peroxide, and myeloperoxidase uses peroxide and chloride to form hypochlorous acid. NADPH oxidase failure causes chronic granulomatous disease. Nitric-oxide synthase uses arginine, oxygen and NADPH to form NO, linking the pathway to vascular signaling.

Sources for this section

Biochemistry Metabolism (Part 1) Notes - Hawraa.pdf - PDF p.1: phosphogluconate pathway mentioned

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - Ch.13 pp.277-297

Dietary lipid digestion: from meal to absorbable products

SOURCE BASIS: lecture + book

Most dietary lipid is TAG. Phospholipids, free cholesterol, cholesteryl esters and fat-soluble vitamins accompany it. Hydrophobic lipid must be presented at a lipid-water interface before water-soluble digestive enzymes can work efficiently. Bile salts emulsify lipid droplets and stabilize smaller particles; they do not hydrolyze ester bonds. Peristalsis supplies mechanical mixing.

Agent	Site/activation	Substrate and result
Lingual and gastric lipases	Work in acidic stomach conditions; important for milk fat and when pancreatic function is poor	Partial TAG hydrolysis, particularly involving shorter fatty-acid chains.
Pancreatic lipase + colipase	Small intestine. Trypsin activates procolipase. Colipase anchors lipase at the interface despite bile salts	Removes TAG fatty acids at positions 1 and 3, leaving 2-monoacylglycerol + two fatty acids.
Cholesterol esterase	Pancreatic secretion; bile salts enhance activity	Cholesteryl ester to free cholesterol + fatty acid.
Phospholipase A2	Pancreatic proenzyme activated by trypsin	Removes fatty acid at phospholipid position 2, producing lysophospholipid; further hydrolysis can follow.
CCK	Intestinal response to lipid and partially digested protein	Gallbladder contraction and pancreatic enzyme secretion; slows gastric emptying.
Secretin	Intestinal response to acidic chyme	Pancreatic bicarbonate secretion creates a suitable pH for pancreatic enzymes.

Do not turn a source generalization into a chemical rule

TAG is glycerol esterified to three fatty acids, not cholesterol plus three fatty acids. Oils and fats differ largely in melting behavior, influenced by saturation and chain length. Plant versus animal origin is not an absolute solid/liquid rule.

Source-specific intake values

The Esmaeili 1.1 transcript, lines 99-100, describes about 78 g dietary lipid per day. Lippincott 6e PDF p.328 describes about 81 g/day for U.S. adults in that edition. Both emphasize more than 90% as TAG. These are attributed population descriptions, not a universal daily requirement, and should not be averaged into an invented value.

Relatively acid-stable lingual and gastric lipases are particularly helpful for short/medium-chain TAG in milk. Their contribution matters in neonates and can partly support lipid digestion when pancreatic function is impaired. This does not mean they fully replace pancreatic digestion of all dietary long-chain fat.

Sources for this section

Lipid metabolism Esmaeili 1.1.transcript.txt - entire transcript

Lipid metabolism Esmaeili 1.2.transcript.txt - entire transcript

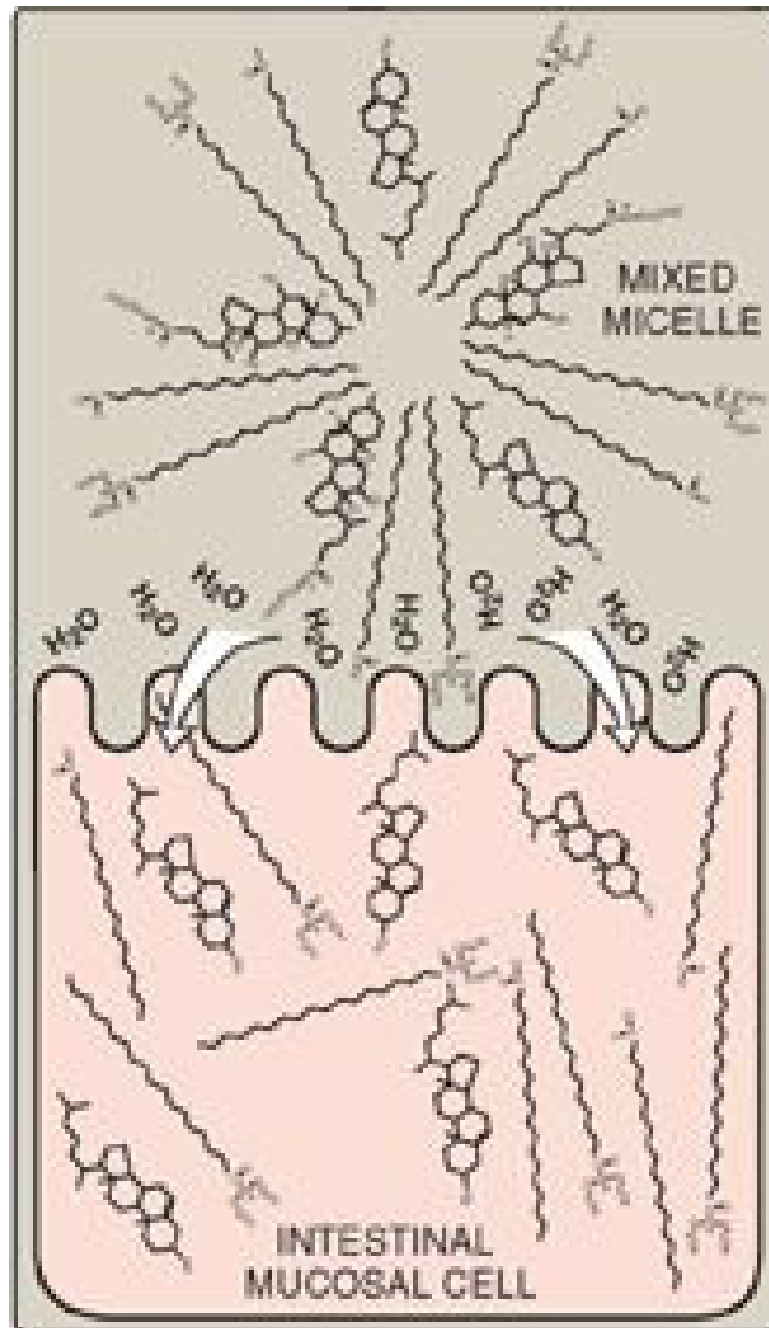
Lipid metabolism1 Esmaeili 1.3.transcript.txt - entire transcript

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - Ch.15 pp.326-331

Micelles, enterocytes, chylomicrons and remnant uptake

SOURCE BASIS: lecture + book

Mixed micelles contain bile salts, fatty acids, 2-monoacylglycerol, cholesterol and fat-soluble vitamins. They facilitate movement through the unstirred aqueous layer to the enterocyte brush border. The lipid contents enter the mucosal cell; the entire micelle is not simply converted into a chylomicron. Most bile salts remain in the lumen until reabsorption later, mainly in terminal ileum.



The source image shows the mixed micelle at the intestinal brush border. Distinguish its luminal transport role from intracellular assembly of a protein-containing chylomicron.

Identify: Mixed micelle; Bile-salt surface; Lipid contents; Brush border; Intestinal mucosal cell

Figure source: Lippincott Biochemistry 6e, PDF p. 333, extracted source diagram

Long-chain fatty acids are activated to acyl-CoA at an ATP-to-AMP cost. Acyltransferases rebuild TAG from 2-monoacylglycerol through DAG. Cholesterol is esterified intracellularly by ACAT. TAG and cholesteryl esters occupy the hydrophobic particle core, while phospholipid, free cholesterol and

apoprotein form the surface. ApoB-48 and microsomal triglyceride transfer protein are required for chylomicron assembly. Chylomicrons leave by exocytosis into lacteals, then lymph and the thoracic duct, reaching venous blood at the left venous angle. Short- and medium-chain fatty acids largely reach portal blood without the micelle/chylomicron route.

In plasma, chylomicrons acquire apoC-II and apoE from HDL. ApoC-II activates lipoprotein lipase on capillary endothelium of adipose and muscle. Released fatty acids enter tissues for storage or oxidation. Glycerol travels especially to liver, whose glycerol kinase can make glycerol-3-phosphate. Adipose has little glycerol-kinase activity and relies heavily on glucose-derived DHAP for the glycerol backbone of storage TAG. After TAG depletion, the cholesterol-rich remnant is taken up by liver through apoE-dependent recognition.

Failure	Immediate consequence	Clinical pattern
Pancreatic insufficiency or blocked ducts	Digestive enzymes fail to reach lipids	Steatorrhea and fat-soluble vitamin malabsorption.
Cholestasis or impaired bile-salt availability	Emulsification/micellar transport inadequate	Fat malabsorption; distinguish from a plasma lipoprotein clearance defect.
LPL or apoC-II deficiency	Chylomicron TAG cannot be efficiently cleared	Fasting chylomicronemia and severe hypertriglyceridemia; pancreatitis risk.
ApoE-related remnant-clearance defect	Remnants accumulate	Type III dysbetalipoproteinemia, involving both chylomicron and VLDL remnants.
ApoB-particle assembly failure	Intestinal lipid export impaired	Fat malabsorption and deficient delivery of fat-soluble vitamins.

CFTR explanation requires tissue specificity

Cystic fibrosis can obstruct pancreatic ducts with thick secretions and cause exocrine insufficiency. CFTR physiology differs between secretory epithelia and sweat ducts: sweat duct failure impairs salt reabsorption, so do not repeat the transcript claim that every affected tissue simply secretes less chloride in the same way.

Sources for this section

Lipid metabolism Esmaeili 1.1.transcript.txt - entire transcript

Lipid metabolism Esmaeili 1.2.transcript.txt - entire transcript

Lipid metabolism1 Esmaeili 1.3.transcript.txt - entire transcript

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - Ch.15 pp.331-337; Ch.18 pp.420-425

Fatty-acid synthesis and TAG storage

SOURCE BASIS: book extension

Fatty-acid synthesis predominates when carbon and energy are abundant. Mitochondrial acetyl-CoA cannot directly cross the inner membrane, so citrate carries its acetyl units to the cytosol. ATP-citrate lyase releases cytosolic acetyl-CoA. Acetyl-CoA carboxylase uses ATP, bicarbonate and biotin to make malonyl-CoA. This is the regulated committed step. Citrate activates ACC, long-chain acyl-CoA inhibits it, and AMPK-mediated phosphorylation inhibits it during energy shortage. Insulin favors activation and increases lipogenic enzyme expression.

Fatty-acid synthase holds the growing chain on its ACP domain using phosphopantetheine derived from vitamin B5. One acetyl primer and seven malonyl units yield palmitate after seven cycles of condensation, reduction, dehydration and reduction. Each cycle consumes two NADPH. Seven ATP form the malonyl-CoA units, and fourteen NADPH supply reducing power, excluding upstream citrate-shuttle costs. The added bicarbonate carbon is released during condensation and is not retained in palmitate. The thioesterase releases free palmitate, not palmitoyl-CoA.

- PPP and cytosolic malic enzyme supply NADPH. Malic enzyme converts malate to pyruvate with oxidative decarboxylation, unlike NADH-linked malate dehydrogenase.
- ER enzymes elongate and desaturate fatty acids. Humans cannot create the omega-6 and omega-3 parent structures de novo: linoleic and alpha-linolenic acids are essential.
- TAG synthesis joins activated fatty acids to glycerol-3-phosphate through phosphatidic acid and DAG. Liver can use glycerol directly and export TAG in VLDL. Adipose predominantly obtains glycerol-3-phosphate from glycolysis and stores TAG.
- Insulin coordinates GLUT4-mediated glucose uptake, glycerol-backbone supply and adipose lipid storage. During fasting, reduced insulin and catecholamine signals favor lipolysis.

Malonyl-CoA separates incompatible pathways

Malonyl-CoA is a synthesis substrate and inhibits CPT-I. Therefore newly synthesized fatty acids are not simultaneously imported into mitochondria for rapid oxidation. The carnitine shuttle, not beta-oxidation chemistry itself, is the controlled entry gate.

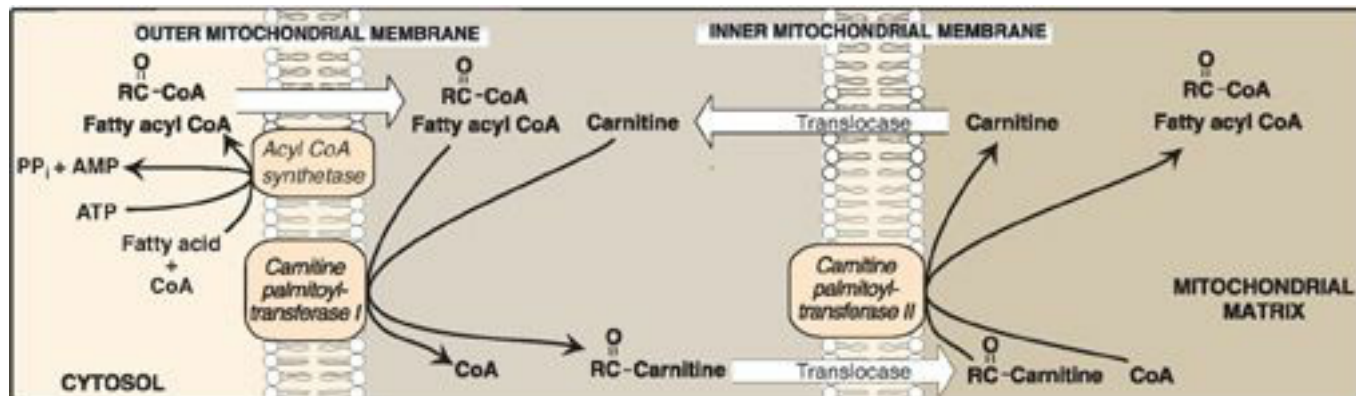
Sources for this section

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - Ch.16 pp.340-355

Lipolysis, carnitine transport and beta-oxidation

SOURCE BASIS: book extension

Adipose TAG mobilization involves sequential lipases, including adipose triglyceride lipase and hormone-sensitive lipase. Released nonesterified fatty acids travel bound to albumin. Activation by acyl-CoA synthetase consumes ATP to AMP plus pyrophosphate, costing two high-energy phosphate equivalents. Long-chain acyl groups cross the inner mitochondrial membrane as acylcarnitine: CPT-I forms acylcarnitine, a translocase exchanges it across the membrane, and CPT-II regenerates acyl-CoA in the matrix.



Carnitine shuttle source diagram. Identify the outer-membrane CPT-I, inner-membrane translocase and matrix-side CPT-II. Acyl groups cross; the original cytosolic CoA does not.

Identify: CPT-I; Acylcarnitine; Translocase; CPT-II; Matrix acyl-CoA

Figure source: Lippincott Biochemistry 6e, PDF p. 357, extracted source diagram

Repeated step	Enzyme/function	Product
First oxidation	Acyl-CoA dehydrogenase	trans-2-enoyl-CoA and FAD-linked reducing equivalents
Hydration	Enoyl-CoA hydratase	L-3-hydroxyacyl-CoA
Second oxidation	3-hydroxyacyl-CoA dehydrogenase	3-ketoacyl-CoA and NADH
Thiolysis	Beta-ketothiolase plus CoA	Acetyl-CoA and acyl-CoA shortened by two carbons

For an even saturated n-carbon fatty acid, n/2 acetyl-CoA units require n/2 minus one beta-oxidation cycles. Palmitate (16:0) gives eight acetyl-CoA, seven NADH and seven FADH₂. At 2.5 ATP/NADH and 1.5 ATP/FADH₂: $8 \times 10 + 7 \times 2.5 + 7 \times 1.5 = 108$ ATP before activation, or 106 net from free palmitate. Starting with palmitoyl-CoA omits that activation charge. The old 6e calculation gives 131 gross and 129 net using 3/2 ratios. State the convention rather than treating these as different carbon pathways.

Special case	Additional requirement	Exam distinction
Odd-chain fatty acid	Propionyl-CoA carboxylase (biotin), epimerase, methylmalonyl-CoA mutase (B12)	Succinyl-CoA can contribute net glucogenic carbon.
Unsaturated fatty acid	Enoyl-CoA isomerase; polyunsaturated substrates may also need NADPH-dependent reductase	Yield is lower than the corresponding saturated chain; specify double-bond arrangement before calculating.
Very-long-chain fatty acid	Initial peroxisomal shortening	Initial oxidation transfers electrons to oxygen, creating H ₂ O ₂ rather than respiratory-chain ATP.
Phytanic acid	Peroxisomal alpha-oxidation	A beta-methyl branch obstructs ordinary beta-oxidation; Refsum disease is the classic defect.
Omega oxidation	ER pathway produces dicarboxylic acids	Can increase when mitochondrial beta-oxidation is impaired.

Hypoketotic hypoglycemia

MCAD deficiency or a hepatic long-chain fatty-acid entry defect impairs fuel oxidation and reduces ketone production during fasting. Glucose demand remains high while hepatic energy supply for gluconeogenesis falls. This differs from the marked ketosis expected in severe insulin deficiency.

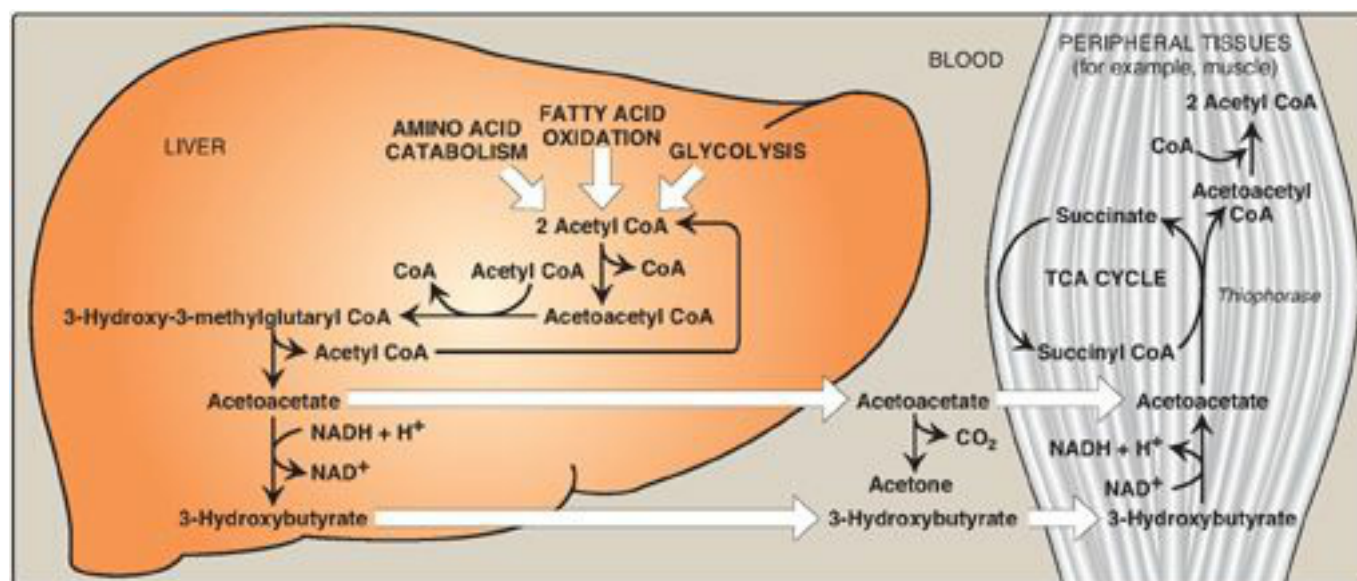
Sources for this section

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - Ch.16 pp.355-363; old ATP accounting p.359

Ketogenesis and ketone utilization

SOURCE BASIS: book extension

During fasting, fatty acids reach liver and generate abundant acetyl-CoA and NADH. Oxaloacetate is diverted toward gluconeogenesis and its reduction to malate is favored by the redox state. Acetyl-CoA therefore exceeds the capacity for complete TCA oxidation. Liver mitochondria condense acetyl-CoA to acetoacetyl-CoA, add another acetyl-CoA through mitochondrial HMG-CoA synthase, then use HMG-CoA lyase to form acetoacetate. Acetoacetate can be reduced to beta-hydroxybutyrate or spontaneously decarboxylate to acetone.



Ketone synthesis in liver and utilization in peripheral mitochondria. Note the separate HMG-CoA pathway and the peripheral thiophorase reaction.

Identify: Mitochondrial HMG-CoA; Acetoacetate; Beta-hydroxybutyrate; Thiophorase/SCOT; Acetoacetyl-CoA

Figure source: Lippincott Biochemistry 6e, PDF p. 366, extracted source diagram

Ketones are water-soluble fuels. Extrahepatic mitochondria oxidize beta-hydroxybutyrate to acetoacetate, then SCOT (thiophorase) transfers CoA from succinyl-CoA. Thiolase generates two acetyl-CoA for oxidation. Liver lacks SCOT and therefore exports rather than consumes ketones. Mature RBCs cannot use them because they lack mitochondria. Brain ketone use increases during prolonged fasting and reduces, but does not abolish, the need for glucose and amino-acid-derived gluconeogenesis.

Three chemically different ketone bodies

Beta-hydroxybutyrate is chemically a hydroxy acid, despite the physiological name ketone body. Acetone does not contribute meaningful ATP. A high hepatic NADH/NAD⁺ ratio favors beta-hydroxybutyrate relative to acetoacetate. Unrestrained ketoacid production can consume bicarbonate and cause high-anion-gap acidosis.

Sources for this section

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - Ch.16 pp.364-369; Ch.24 pp.600-608

Membrane lipids, sphingolipidoses and eicosanoids

SOURCE BASIS: book extension

Class	Structural logic	Function
Glycerophospholipid	Glycerol with fatty acids and phosphate-linked head group	Membrane structure, signaling and surfactant.
Sphingomyelin	Ceramide plus phosphocholine	Myelin and membrane phospholipid, not glycerol based.
Cerebroside	Ceramide plus one sugar	Neutral glycosphingolipid.
Ganglioside	Ceramide plus oligosaccharide containing sialic acid	Negatively charged neuronal membrane glycolipid.
Cardiolipin	Diphosphatidylglycerol	Inner mitochondrial membrane organization.
Dipalmitoylphosphatidylcholine	Saturated phosphatidylcholine species	Important pulmonary surfactant lipid.

Phospholipase A2 releases the fatty acid at glycerophospholipid position 2, often arachidonate. Phospholipase C cleaves PIP2 into DAG and IP3: DAG remains membrane associated and activates protein kinase C, while IP3 releases stored intracellular calcium. A GPI anchor attaches selected proteins to the external membrane surface. Loss of protective GPI-anchored complement regulators on blood cells contributes to paroxysmal nocturnal hemoglobinuria.

Disease	Enzyme / accumulated substrate	Distinguishing clue
Tay-Sachs	Hexosaminidase A / GM2 ganglioside	Progressive neurodegeneration; classic absence of hepatosplenomegaly.
Gaucher	Glucocerebrosidase / glucocerebroside	Macrophage storage, organ enlargement and bone disease.
Niemann-Pick A/B	Sphingomyelinase / sphingomyelin	Sphingomyelin storage; type C instead concerns intracellular cholesterol trafficking.

Disease	Enzyme / accumulated substrate	Distinguishing clue
Fabry	Alpha-galactosidase A / globotriaosylceramide	X-linked exception; painful neuropathy, angiokeratomas and renal/cardiac disease.
Krabbe	Galactocerebrosidase	Demyelination with toxic psychosine accumulation.
Metachromatic leukodystrophy	Arylsulfatase A / sulfatides	Central and peripheral demyelination.

Arachidonic acid can enter cyclooxygenase pathways to prostaglandins and thromboxanes or lipoxygenase pathways to leukotrienes. PGI₂ generally supports vasodilation and inhibits platelet aggregation; TXA₂ favors aggregation and vasoconstriction. LTB₄ recruits leukocytes, while cysteinyl leukotrienes promote bronchoconstriction and vascular permeability. Aspirin irreversibly inhibits existing platelet COX-1, but does not block leukotriene synthesis. Its antiplatelet effect persists across the affected platelet lifespan and recovery is mainly through production of new platelets, unlike nucleated endothelium replenishing its enzyme. Do not turn this into the absolute claim that platelets cannot make any protein: they retain translational machinery.

Sources for this section

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - Ch.17 pp.374-405

Cholesterol, bile acids and steroid synthesis

SOURCE BASIS: book extension

Cholesterol has a steroid nucleus and a C3 hydroxyl group. Esterification of that hydroxyl produces a more hydrophobic storage/transport form. Carbon comes from acetyl-CoA and reducing power from NADPH. Cytosolic/ER cholesterol synthesis proceeds through HMG-CoA, mevalonate, activated isoprenes, squalene and lanosterol. HMG-CoA reductase makes mevalonate and is a key regulated step. It must be distinguished from mitochondrial HMG-CoA synthase/lyase in ketogenesis.

- Low cellular cholesterol allows SREBP processing and increased expression of HMG-CoA reductase and LDL receptors. High sterol availability suppresses this program and promotes reductase degradation.
- AMPK phosphorylation inhibits reductase. Insulin-associated dephosphorylation favors activity. Statins competitively inhibit HMG-CoA reductase, increasing hepatic demand for LDL-derived cholesterol.
- Humans do not completely oxidize the cholesterol ring to CO₂ for energy. Elimination is through free cholesterol in bile and conversion to bile acids. Cholesterol 7- α -hydroxylase is a regulated step in the classical bile-acid pathway.
- Primary bile acids are cholic and chenodeoxycholic acids. Conjugation with glycine or taurine improves detergent function. Intestinal bacteria generate secondary bile acids including deoxycholic and lithocholic acids.
- Efficient ileal reabsorption and portal return create the enterohepatic circulation. Loss of ileal absorption can reduce the bile-salt pool and compromise fat absorption. Cholesterol gallstones form when bile cannot keep cholesterol soluble.

Steroid synthesis begins with cholesterol delivery into mitochondria and side-chain cleavage to pregnenolone. Tissue-specific enzyme expression then directs production toward glucocorticoids,

mineralocorticoids or sex steroids. Steroid hormones generally bind intracellular receptors and alter transcription. Deficient steroid enzymes alter both downstream hormone production and upstream precursors, so clinical questions should be reasoned through the blocked step rather than memorized as an isolated name.

Sources for this section

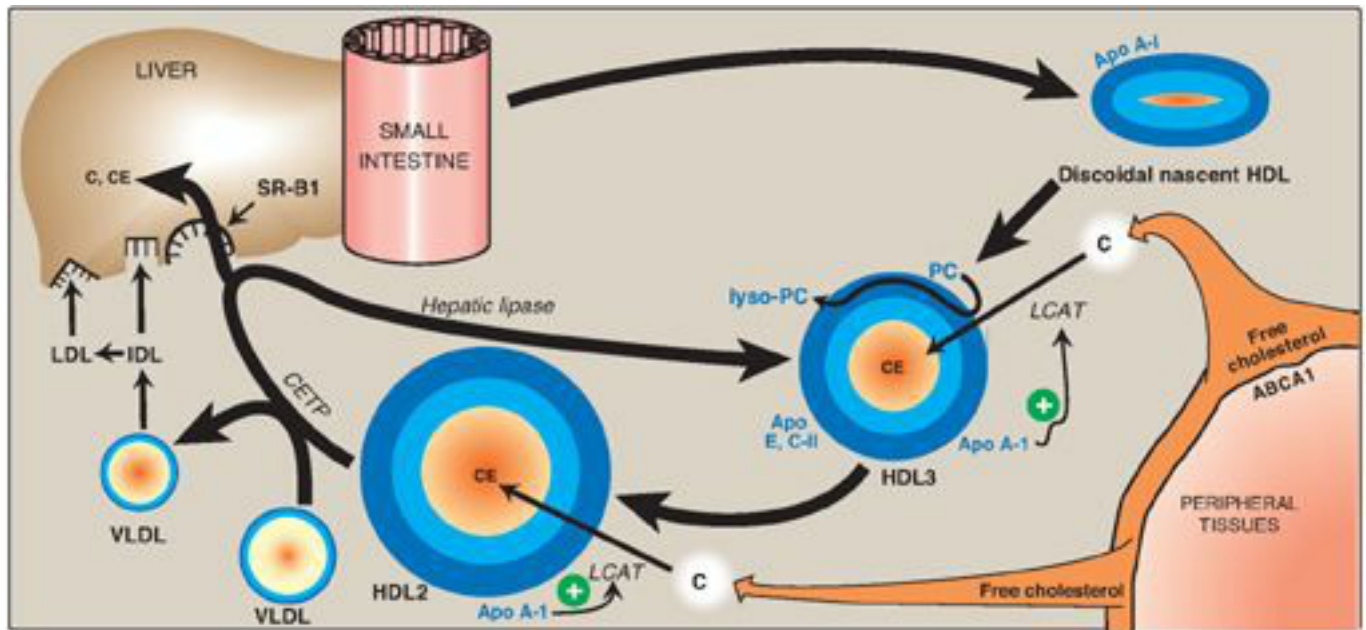
Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - Ch.18 pp.406-419,434-442

Lipoproteins and apoproteins: an integrated transport map

SOURCE BASIS: lecture + book

Particle	Origin and principal job	Recognition / fate
Chylomicron	Intestine; dietary TAG transport	ApoB-48 structure, apoC-II activates LPL, apoE directs remnant uptake.
VLDL	Liver; export of endogenous TAG	ApoB-100; LPL removes TAG to form IDL.
IDL	VLDL remnant	ApoE-dependent liver uptake or further lipolysis to LDL.
LDL	Cholesterol-rich product of VLDL/IDL metabolism	ApoB-100 binds LDL receptor; endocytosis delivers cholesterol.
HDL	Liver/intestine apoA-I, acquiring lipid in plasma	Cholesterol efflux, apoC-II/apoE reservoir, reverse cholesterol transport.

Factor	Function	Do not confuse with
ApoA-I	Activates plasma LCAT	ApoC-II activation of LPL
ApoB-48	Structural protein of chylomicrons	ApoB-100 in VLDL/IDL/LDL
ApoB-100	Structural protein and LDL-receptor ligand	ApoE-mediated remnant recognition
ApoC-II	LPL activator	Pancreatic colipase
ApoE	Remnant receptor ligand	ApoB-48, which does not substitute as the LDL ligand
LCAT	Esterifies cholesterol on HDL using phosphatidylcholine	Intracellular ACAT, which uses acyl-CoA
CETP	Exchanges cholesteryl ester and TAG among HDL and apoB particles	A cholesterol-synthesis enzyme
ABCA1 / SR-BI	Cellular cholesterol efflux to apoA-I / selective hepatic HDL lipid uptake	LDL receptor whole-particle endocytosis



HDL source figure: follow peripheral cholesterol acquisition, LCAT esterification, CETP exchange and delivery to liver.
HDL-C concentration alone is not equivalent to all HDL functions.

Identify: ApoA-I; ABCA1; LCAT; CETP; SR-BI

Figure source: Lippincott Biochemistry 6e, PDF p. 432, extracted source diagram

LDL-receptor defects or impaired apoB binding reduce LDL clearance and increase circulating LDL. Macrophages take up modified LDL through scavenger mechanisms that do not provide the same feedback-limited clearance as the LDL receptor, promoting foam cells. Lp(a) is an LDL-like particle with apo(a) attached to apoB-100. Severe chylomicronemia primarily warns of pancreatitis, whereas prolonged exposure to atherogenic apoB particles drives arterial risk.

Sources for this section

Lipid metabolism Esmaeili 1.1.transcript.txt - entire transcript

Lipid metabolism Esmaeili 1.2.transcript.txt - entire transcript

Lipid metabolism1 Esmaeili 1.3.transcript.txt - entire transcript

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - Ch.18 pp.420-442

Protein digestion, amino-acid transport and nitrogen collection

SOURCE BASIS: book extension

Protein digestion starts with acid denaturation and pepsin action in the stomach. Pancreatic proteases are secreted as zymogens. Enteropeptidase initiates trypsin activation at the intestinal surface, and trypsin activates other zymogens. Brush-border and intracellular peptidases complete digestion. Amino acids and small peptides use specific transport systems; transport defects can produce selective urinary losses even when metabolism inside the cell is intact.

Body proteins continually turn over. Ubiquitin tagging directs many selected intracellular proteins to the proteasome, whereas lysosomes degrade endocytosed material and proteins delivered through autophagy. There is no dedicated storage polymer for surplus amino acids. Their nitrogen must be disposed of while carbon skeletons enter energy metabolism or synthesis.

Reaction/process	Meaning	Key example
Transamination	Transfers an amino group to an alpha-keto acid without releasing free ammonia	ALT: alanine + alpha-ketoglutarate gives pyruvate + glutamate.
AST reaction	Transfers glutamate nitrogen to oxaloacetate	Produces aspartate for the second nitrogen of urea.
Oxidative deamination	Glutamate dehydrogenase releases ammonia and regenerates alpha-ketoglutarate	Mitochondrial reaction links nitrogen collection to urea synthesis.
Glutamine formation	Glutamine synthetase captures ammonia using ATP	Safe nitrogen transport from tissues.
Glutaminase	Releases glutamine amide nitrogen as ammonia	Liver disposal and renal ammonium handling.
Glucose-alanine cycle	Muscle transfers amino nitrogen to pyruvate, exports alanine	Liver uses pyruvate for glucose and nitrogen for urea.

PLP versus NAD(P)+

Aminotransferases require PLP (B6). Glutamate dehydrogenase uses a nicotinamide coenzyme. A transaminase does not consume ATP to remove nitrogen as ammonia; it redistributes nitrogen to another carbon skeleton.

Sources for this section

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - Ch.19 pp.448-463

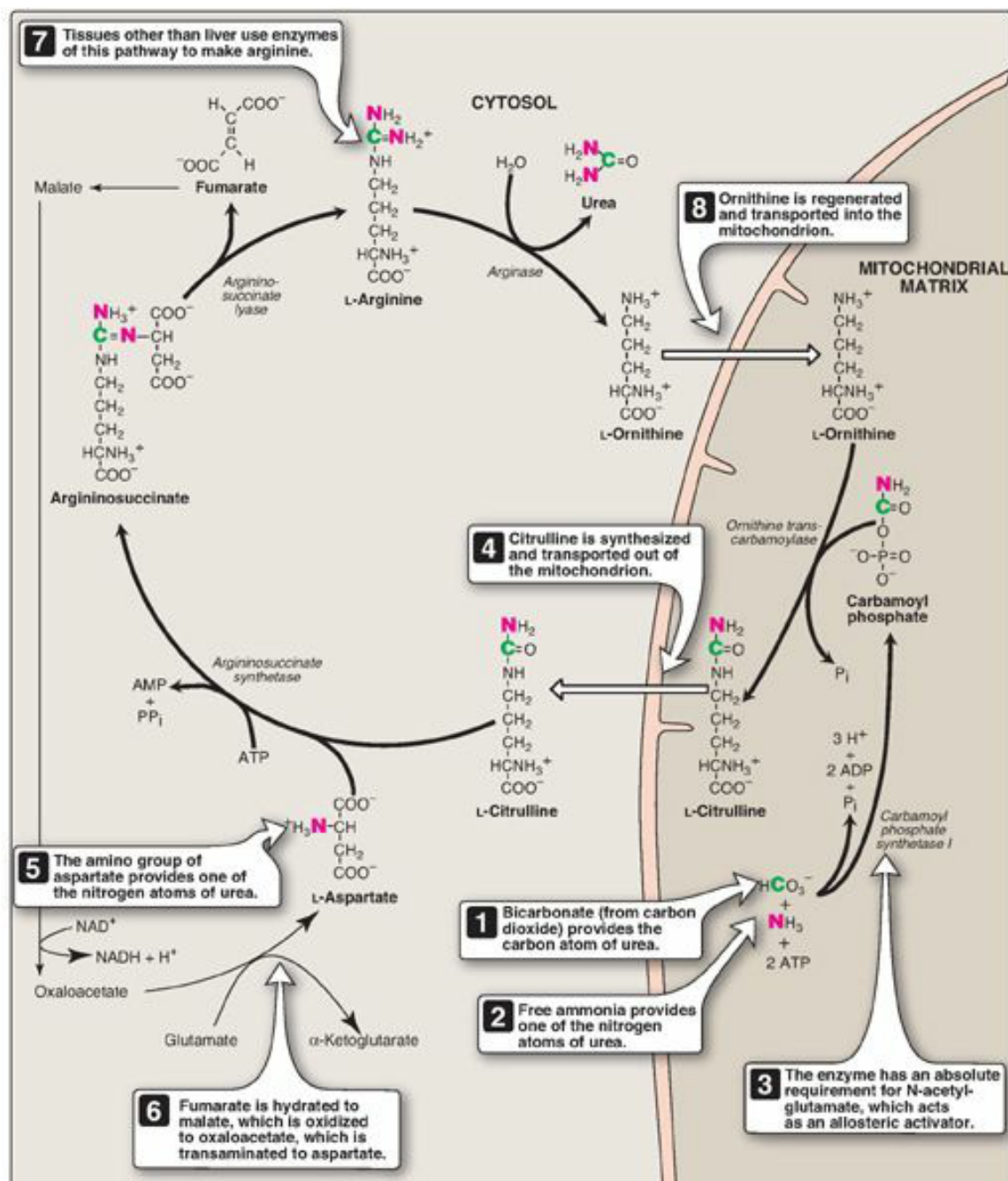
Urea cycle: atom origins, compartments and diagnostic contrasts

SOURCE BASIS: book extension

The liver converts ammonia to urea for renal excretion. Urea receives one nitrogen from free ammonia, one from aspartate, and its carbon from bicarbonate. The first two reactions occur in mitochondrial matrix; later reactions occur in cytosol. Ornithine and citrulline transport joins the compartments.

Enzyme	Reaction	Energy / control
CPS-I	Ammonia + bicarbonate to carbamoyl phosphate	Consumes two ATP; requires N-acetylglutamate.
Ornithine transcarbamylase	Ornithine + carbamoyl phosphate to citrulline	Mitochondrial; citrulline exported.
Argininosuccinate synthetase	Citrulline + aspartate to argininosuccinate	ATP to AMP + PPi, costing two high-energy bonds.

Enzyme	Reaction	Energy / control
Argininosuccinate lyase	Argininosuccinate to arginine + fumarate	Fumarate connects to malate/oxaloacetate and aspartate.
Arginase	Arginine + water to ornithine + urea	Ornithine recycled; urea released.



Source urea-cycle figure, showing both compartments and the origin of each nitrogen atom. Follow transport as well as chemistry.

Identify: CPS-I; OTC; Citrulline; Aspartate; Argininosuccinate; Arginine; Ornithine; Fumarate

Figure source: Lippincott Biochemistry 6e, PDF p. 465, extracted source diagram

Three ATP molecules are consumed, but four high-energy phosphate bonds are spent. N-acetylglutamate synthase uses acetyl-CoA and glutamate and is stimulated by arginine. Increased nitrogen intake or catabolism increases cycle demand. Hyperammonemia disrupts brain metabolism and astrocyte glutamine handling, producing neurological dysfunction; no single ammonia level alone identifies the enzyme defect.

Contrast	CPS-I / urea cycle	CPS-II / pyrimidine synthesis
Location	Mitochondria	Cytosol
Nitrogen donor	Free ammonia	Glutamine
Activator / feedback	N-acetylglutamate required	PRPP activates; UTP inhibits
Defect clue	Hyperammonemia with low citrulline in an early block	Different nucleotide synthesis phenotype

OTC versus CPS-I deficiency

OTC deficiency is X-linked and can divert excess carbamoyl phosphate into pyrimidine synthesis, raising urinary orotic acid. CPS-I deficiency blocks carbamoyl phosphate formation and typically lacks that orotic-acid rise. Both can cause hyperammonemia; primary UMP-synthase deficiency causes orotic aciduria without a primary urea-cycle block.

Sources for this section

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - Ch.19 pp.464-474; Ch.22 pp.551-555

Amino-acid carbon skeletons, essentiality and one-carbon metabolism

SOURCE BASIS: book extension

Glucogenic amino acids yield pyruvate or TCA intermediates that can provide net glucose. Ketogenic amino acids yield acetyl-CoA or acetoacetate precursors. Leucine and lysine are exclusively ketogenic. Phenylalanine, tyrosine, tryptophan and isoleucine have both routes. Threonine classification differs among teaching schemes: the inspected 6th edition emphasizes its glucogenic route. Know the specific reaction shown rather than assuming a list copied from another edition is identical.

Entry product	Representative amino acids	Useful link
Pyruvate	Alanine, serine, glycine, cysteine; portions of tryptophan metabolism	Gluconeogenesis through pyruvate carboxylase.
Oxaloacetate	Aspartate, asparagine	Direct connection to AST and urea nitrogen.
Alpha-ketoglutarate	Glutamate, glutamine, proline, arginine, histidine	Nitrogen collection and TCA replenishment.
Succinyl-CoA	Methionine, valine, isoleucine, threonine pathways	Propionyl-CoA, biotin and B12 link.
Fumarate	Phenylalanine and tyrosine	Their other carbon fragments are ketogenic.
Acetyl-CoA/acetoacetate	Leucine, lysine and ketogenic portions of mixed amino acids	No net conversion of acetyl carbon into glucose.

The essential adult amino acids are histidine, isoleucine, leucine, lysine, methionine, phenylalanine, threonine, tryptophan and valine. Tyrosine becomes conditionally essential when phenylalanine cannot be hydroxylated. Cysteine depends on an adequate sulfur supply from methionine. Branched-chain amino acids are initially metabolized substantially in peripheral tissues, especially muscle; the liver is not the sole site of all amino-acid catabolism.

Methionine plus ATP forms S-adenosylmethionine, a major methyl donor. After methyl transfer, S-adenosylhomocysteine yields homocysteine. Remethylation to methionine requires methyl-B12 and methyl-THF. Transsulfuration combines homocysteine with serine via B6-dependent cystathionine beta-synthase and ultimately yields cysteine. THF carries one-carbon units at different oxidation states for purines, thymidylate and selected amino-acid reactions. Biotin carries CO₂ in carboxylation, not the same one-carbon pool.

Carbon classification in the inspected 6e	Amino acids	Reasoning
Exclusively ketogenic	Leucine, lysine	Yield acetyl/acetoacetyl carbon without a net glucogenic product.
Both glucogenic and ketogenic	Isoleucine, phenylalanine, tyrosine, tryptophan	Generate a glucogenic entry intermediate plus acetyl-CoA or acetoacetate.
Glucogenic in this 6e table	Alanine, arginine, asparagine, aspartate, cysteine, glutamate, glutamine, glycine, histidine, methionine, proline, serine, threonine, valine	Produce pyruvate or a net TCA intermediate.

Threonine classification depends on the represented route

The inspected 6e Figure 20.2 lists threonine as glucogenic and emphasizes its alpha-ketobutyrate/propionyl-CoA route. Some teaching lists classify it as both by including acetyl-CoA-producing routes; the source summary Figure 20.14 itself depicts such an alternate route faintly. For an exam, identify the route or the named source rather than presenting one list as a universal contradiction-free rule.

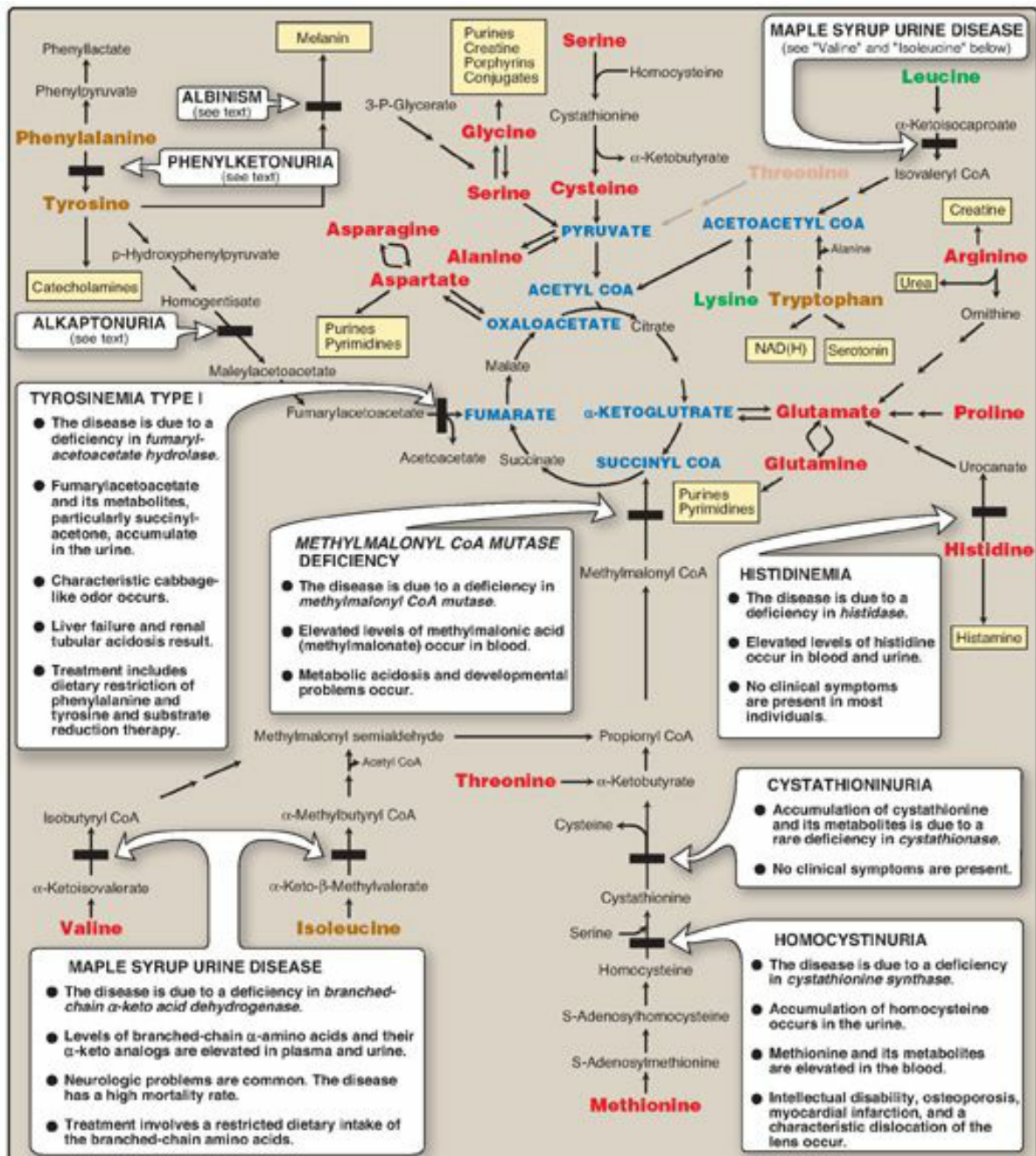


Figure 20.14 from the local book maps amino-acid carbon entry, special products and inherited blocks. The adjacent table states the edition-specific threonine convention; text below explains the tyrosinemia type I box.

Identify: Pyruvate; Oxaloacetate; Alpha-ketoglutarate; Succinyl-CoA; Fumarate; Acetyl-CoA; Acetoacetyl-CoA

Figure source: Lippincott Biochemistry 6e, PDF p. 493, extracted source diagram

Sources for this section

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - Ch.20 pp.477-493,501-502

Amino-acid disorders: reason from the blocked reaction

SOURCE BASIS: book extension

Condition	Block	Reasoning pattern
Classic PKU	Phenylalanine hydroxylase	High phenylalanine, low tyrosine; neurological toxicity and reduced pigment.
BH4-related hyperphenylalaninemia	BH4 synthesis or regeneration	Also affects tyrosine and tryptophan hydroxylases, so neurotransmitter deficits accompany elevated phenylalanine.
Maple syrup urine disease	Branched-chain alpha-ketoacid dehydrogenase	Leucine, isoleucine and valine-derived metabolites accumulate; infantile encephalopathy and characteristic odor.
Classic homocystinuria	Cystathionine beta-synthase	High homocysteine and often methionine; thrombosis, lens displacement and skeletal phenotype.
B12 or remethylation defect	Homocysteine to methionine impaired	Homocysteine rises but methionine need not rise; methylmalonate helps identify B12 involvement.
Alkaptonuria	Homogentisate oxidase	Urine darkens on standing, ochronosis and large-joint disease.
Tyrosinase-related albinism	Melanin synthesis	Pigment/visual phenotype without the phenylalanine accumulation of PKU.
Methylmalonyl-CoA mutase or B12 defect	Methylmalonyl-CoA to succinyl-CoA	Methylmalonic acid accumulation and metabolic dysfunction.

Clinical dietary management of inborn errors must preserve essential nutrient requirements while limiting toxic accumulation; it is not simply complete removal of all protein. Maternal hyperphenylalaninemia can harm fetal development even if the fetus does not have PKU. For examination reasoning, separate substrate accumulation, product deficiency and secondary pathway diversion. These summaries are learning material, not individual treatment protocols.

Tyrosinemia type I is deficiency of fumarylacetoacetate hydrolase, the terminal tyrosine-degradation step. Fumarylacetoacetate and toxic metabolites including succinylacetone accumulate; hepatic failure and renal tubular dysfunction are characteristic. Succinylacetone is a particularly useful biochemical discriminator from PKU or alkaptonuria. This mechanism is shown in the source Figure 20.14 at PDF p.493. Histidase deficiency is another named block in that figure and often has no major clinical phenotype; do not assume that every elevated amino acid causes the same severe neonatal syndrome.

Sources for this section

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - Ch.20 pp.494-504

Heme synthesis, porphyrias and bilirubin handling

SOURCE BASIS: book extension

Heme synthesis begins in mitochondria when glycine and succinyl-CoA form ALA through PLP-dependent ALA synthase. Several intermediate steps occur in cytosol before the pathway returns to mitochondria. ALA dehydratase forms porphobilinogen, successive condensation/cyclization steps build the porphyrin ring, and ferrochelatase inserts ferrous iron into protoporphyrin IX. Hepatic ALAS1 responds to heme feedback; erythroid ALAS2 is strongly linked to iron availability. Lead inhibits ALA dehydratase and ferrochelatase.

Porphyrias differ by the accumulating intermediate. Early precursor accumulation can cause neurovisceral attacks, whereas later porphyrin accumulation can create photosensitivity. The name of the enzyme alone is insufficient: identify whether precursors or light-reactive porphyrins accumulate. Heme breakdown by heme oxygenase produces biliverdin, carbon monoxide and iron. Biliverdin reductase forms bilirubin. Unconjugated bilirubin travels bound to albumin; hepatic UDP-glucuronosyltransferase conjugates it, enabling biliary secretion.

Pattern	Serum/urine reasoning	Mechanism
Hemolysis	Predominantly unconjugated bilirubin; no unconjugated bilirubin in urine	Increased heme load exceeds conjugation capacity; albumin-bound pigment is not filtered normally.
Obstruction	Conjugated bilirubin can appear in urine; reduced pigment reaching bowel	Biliary outflow impaired; pale stools can occur.
Hepatocellular injury	Mixed patterns possible	Both conjugation and secretion can be impaired.
Neonatal low conjugation capacity	Unconjugated hyperbilirubinemia	Developmental enzyme limitation must be distinguished from obstruction.

Two different conjugation systems

Bilirubin conjugation uses glucuronic acid. Bile acids are conjugated with glycine or taurine. Cholesterol esterification adds a fatty acid. These are distinct reactions involving different substrates and enzymes.

Porphyria	Deficient enzyme	Useful accumulation/pattern contrast
Acute intermittent porphyria	Hydroxymethylbilane synthase / porphobilinogen deaminase	ALA and PBG rise; neurovisceral attacks without the characteristic porphyrin photosensitivity.
Porphyria cutanea tarda	Uroporphyrinogen decarboxylase	Uroporphyrin-related accumulation; predominantly cutaneous photosensitivity and liver/iron-related associations.
Hereditary coproporphyria	Coproporphyrinogen oxidase	Coproporphyrin-related accumulation; acute attacks may coexist with photosensitivity.

Porphyria	Deficient enzyme	Useful accumulation/pattern contrast
Variegate porphyria	Protoporphyrinogen oxidase	Protoporphyrinogen and upstream metabolites accumulate; acute and cutaneous manifestations can coexist.
Erythropoietic protoporphyria	Ferrochelatase in the classic book pattern	Protoporphyrin accumulation in erythrocytes/plasma and hepatobiliary consequences; photosensitivity.
Congenital erythropoietic porphyria	Uroporphyrinogen III synthase	Type I porphyrin production/accumulation; early photosensitivity.

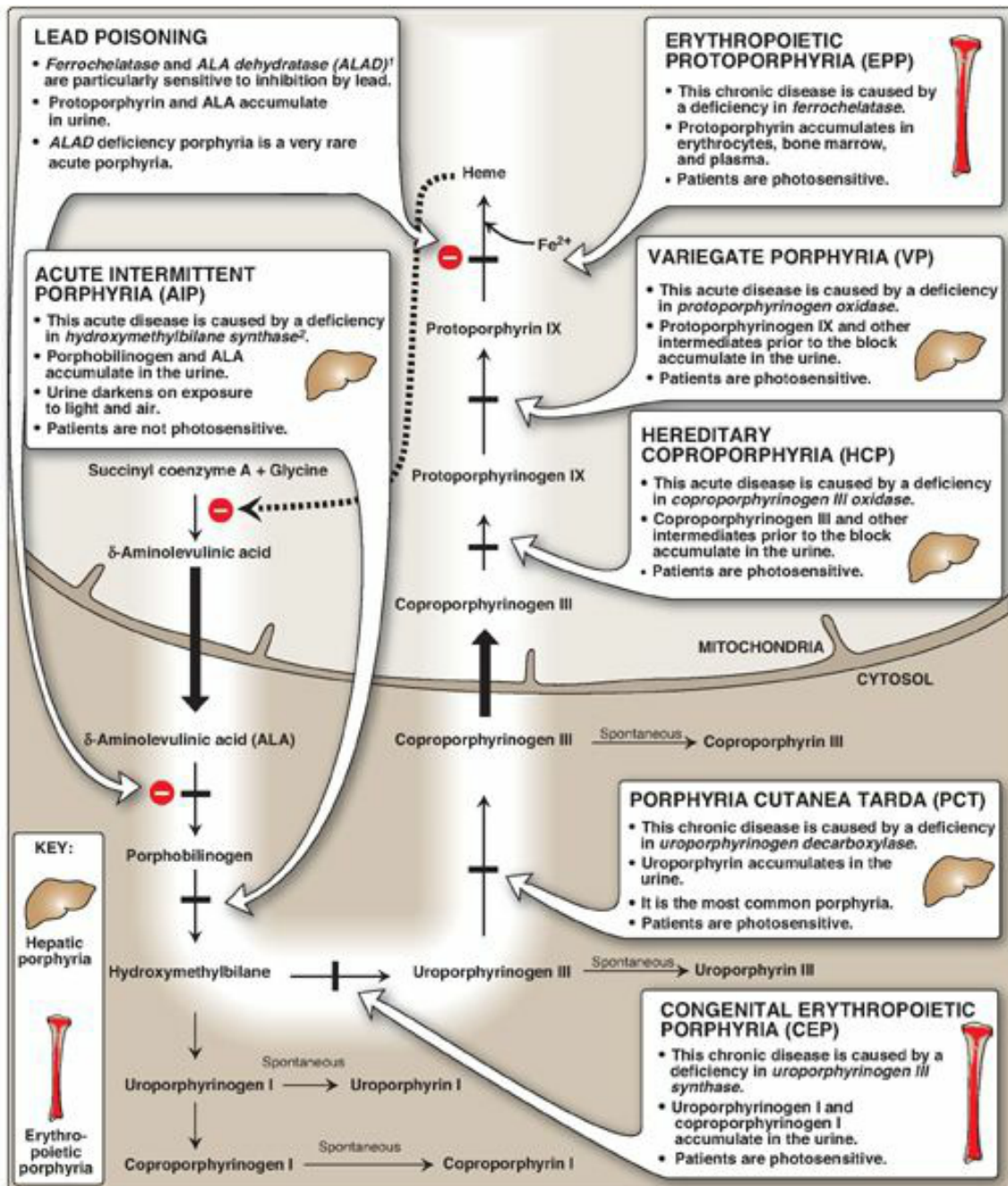


Figure 21.8, local Lippincott PDF p.515 (caption begins p.514): mitochondrial/cytosolic heme pathway and porphyria blocks. Distinguish precursor accumulation from mature heme deficiency.

Identify: ALA synthase; ALA dehydratase; Hydroxymethylbilane synthase; Uroporphyrinogen III synthase; Uroporphyrinogen decarboxylase; Coproporphyrinogen oxidase; Protoporphyrinogen oxidase; Ferrochelatase

Figure source: Lippincott Biochemistry 6e, PDF p. 515, extracted source diagram

A pathway block does not mean all tests rise equally

Acute precursor-producing disorders and photosensitizing porphyrin accumulation are related but distinct patterns. Lead inhibits ALA dehydratase and ferrochelatase; it is an acquired toxic block, not automatically one inherited porphyria. ALAS2 deficiency causes sideroblastic anemia rather than an ordinary porphyria phenotype. This review does not turn the older book treatment paragraph into current management advice.

Sources for this section

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - Ch.21 pp.506-529

Amino acids as precursors of signaling and storage molecules

SOURCE BASIS: book extension

Precursor	Product sequence	Cofactor / significance
Tyrosine	DOPA, dopamine, norepinephrine, epinephrine	Tyrosine hydroxylase uses BH ₄ ; decarboxylation uses PLP; epinephrine methylation uses SAM.
Tryptophan	5-hydroxytryptophan, serotonin, melatonin	Hydroxylation uses BH ₄ and decarboxylation uses PLP.
Histidine	Histamine	PLP-dependent decarboxylation.
Glutamate	GABA	PLP-dependent glutamate decarboxylase.
Arginine	Nitric oxide and citrulline	NOS uses oxygen and reducing power; local vascular and neural signaling.
Glycine + arginine + methyl group from methionine	Creatine, then phosphocreatine	Creatine kinase buffers ATP demand; spontaneous creatinine formation links muscle pool to renal lab measurements.
Glycine, cysteine, glutamate	Glutathione	Antioxidant tripeptide; reduced and oxidized pools must be distinguished.
Tyrosine	Melanin	Copper-dependent tyrosinase, not phenylalanine hydroxylase.

Amino-acid chemistry therefore explains more than fuel oxidation. A deficiency of a shared cofactor can affect neurotransmission, heme synthesis and transamination together. SAM methyl-transfer reactions and THF/B12-dependent remethylation link one-carbon metabolism to these specialized products.

Sources for this section

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - Ch.21 pp.520-529

Purines, pyrimidines, salvage and DNA precursor supply

SOURCE BASIS: book extension

A nucleoside contains a base and sugar; a nucleotide adds phosphate. Purines are built on a pre-existing ribose-phosphate scaffold supplied by PRPP. The committed purine step is glutamine-PRPP amidotransferase, stimulated by substrate availability and inhibited by end products. IMP branches toward AMP or GMP, with reciprocal use of GTP and ATP helping balance their production. One-carbon units from formyl-THF participate in purine-ring construction.

Purine salvage attaches bases to PRPP through APRT or HGPRT, conserving energy. Purine degradation ends in uric acid in humans. Xanthine oxidase acts in the hypoxanthine-to-xanthine-to-urate sequence. HGPRT deficiency reduces salvage and removes

feedback restraint on de novo synthesis; severe deficiency causes Lesch-Nyhan syndrome. Hyperuricemia can result from excess production or impaired excretion and is not by itself identical to symptomatic gout.

Pyrimidine rings are built before attachment to ribose phosphate. Cytosolic CPS-II supplies carbamoyl phosphate using glutamine. The ring proceeds through orotate, which becomes OMP and UMP through UMP-synthase activities. A primary UMP-synthase defect causes orotic aciduria with impaired nucleotide supply, unlike OTC deficiency where orotate rises secondarily to a urea-cycle block.

Enzyme	Function	High-yield association
Ribonucleotide reductase	Converts ribonucleoside diphosphates to deoxy forms	Overall inhibition by dATP; hydroxyurea target.
Thymidylate synthase	dUMP to dTMP using methylene-THF	5-fluorouracil-derived inhibitor; folate becomes DHF.
Dihydrofolate reductase	Regenerates THF from DHF using NADPH	Methotrexate target; needed to sustain nucleotide synthesis.
Adenosine deaminase	Purine breakdown	Deficiency raises deoxyadenosine/dATP, impairing lymphocyte DNA synthesis and causing SCID.
CPS-II	First regulated pyrimidine step	UTP inhibits; PRPP activates; cytosolic and glutamine dependent.

Sources for this section

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - Ch.22 pp.532-559

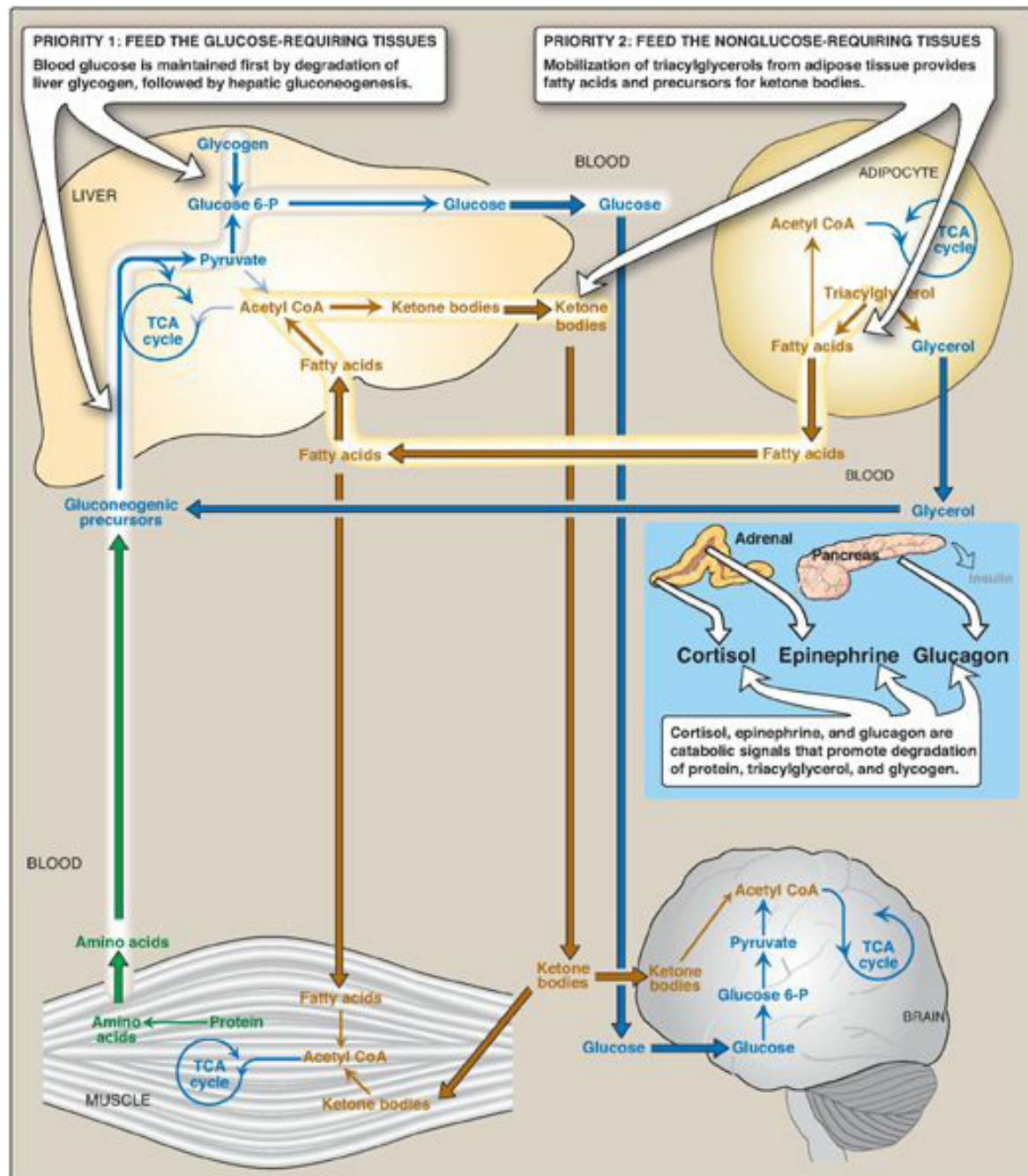
Insulin, glucagon and the fed-fast cycle

SOURCE BASIS: book extension

Pancreatic beta cells process proinsulin to insulin and C-peptide. Glucose metabolism raises ATP, closes ATP-sensitive K⁺ channels, depolarizes the cell and permits calcium entry that triggers secretion. Insulin promotes glucose uptake in muscle/adipose, glycogen synthesis, lipogenesis and protein synthesis. Glucagon from alpha cells primarily coordinates hepatic glucose output and ketogenesis. Amino acids can stimulate both hormones after a protein-containing meal, helping avoid a fall in blood glucose as amino acids are taken up.

Tissue	Fed state	Fasting / prolonged fasting
Liver	Glycogen replenishment, glycolysis, fatty-acid/TAG synthesis and VLDL export	Early glycogenolysis, then increasing gluconeogenesis; fatty-acid oxidation supplies energy and ketones.
Adipose	GLUT4 uptake, TAG synthesis and storage	TAG lipolysis exports fatty acids and glycerol.
Skeletal muscle	Glucose uptake, glycogen and protein synthesis	Fatty acids and ketones support energy; early protein release supplies gluconeogenic amino acids.

Tissue	Fed state	Fasting / prolonged fasting
Brain	Predominantly glucose	Increasing ketone use over prolonged fasting reduces glucose requirement.
RBC	Glucose to lactate throughout	Still needs glucose and PPP NADPH; cannot use fatty acids or ketones.
Kidney cortex	Oxidative metabolism and routine acid-base handling	Increased gluconeogenesis and ammonium excretion in prolonged fasting.



Book concept figure linking tissues during nutrient deprivation. Use the table to distinguish early fasting from the later adaptation that spares protein.

Identify: Liver; Adipose tissue; Muscle; Brain; Glucose; Fatty acids; Ketones

Figure source: Lippincott Biochemistry 6e, PDF p. 609, extracted source diagram

The purpose of adaptation is not to stop glucose production, but to reserve glucose for tissues that require it while making alternative fuel available. As the brain uses more ketones, muscle protein breakdown can diminish. In exercise, ATP demand rises rapidly, calcium helps coordinate glycogen breakdown and TCA oxidation, and muscle contraction can increase glucose uptake independently of insulin.

Sources for this section

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - Chs.23-24 pp.560-613

Diabetes and metabolic perturbations: causal review

SOURCE BASIS: book extension

Type 1 diabetes typically involves profound insulin deficiency. Hepatic glucose output is insufficiently suppressed while peripheral insulin-dependent glucose use falls. Adipose lipolysis delivers fatty acids to liver, promoting ketogenesis. Type 2 diabetes combines insulin resistance with inadequate beta-cell compensation; residual insulin often restrains ketogenesis, but ketoacidosis is not impossible. These categories describe mechanisms, not an inflexible rule based only on age or body size.

- Hyperglycemia can cause osmotic diuresis and water/electrolyte loss. Serum potassium may be normal or high in ketoacidosis despite total-body potassium depletion, because distribution and loss are separate variables.
- Chronic glucose excess promotes nonenzymatic glycation and oxidative/inflammatory injury. HbA1c reflects prior exposure but is also affected by RBC lifespan and hemoglobin-related factors.
- Aldose-reductase flux can consume NADPH and accumulate sorbitol in susceptible tissues. This is one mechanistic link between hyperglycemia and cellular stress, not the sole explanation of every diabetic complication.
- Visceral adipose dysfunction, excess fatty-acid delivery and altered adipokines contribute to insulin resistance. Leptin signals energy stores, whereas ghrelin promotes hunger; common obesity is not generally simple absence of leptin.
- High NADH from ethanol favors lactate formation and TAG accumulation and suppresses gluconeogenesis. Combine nutritional state, glycogen reserve and redox effects when solving an alcohol-metabolism vignette.

Check yourself: Why can fasting liver oxidize fatty acids while simultaneously making glucose?

Fatty-acid oxidation supplies the ATP and reducing power required for gluconeogenesis. Acetyl-CoA activates pyruvate carboxylase and inhibits PDH, directing pyruvate toward oxaloacetate rather than oxidation. Low malonyl-CoA permits CPT-I-mediated fatty-acid entry. The pathways are coordinated rather than contradictory.

Sources for this section

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - Chs.23,25,26 pp.575-579,614-650

Vitamin and cofactor integration for metabolism

SOURCE BASIS: book extension

Vitamin/cofactor	Reactions to recognize	Deficiency / distinction
B1, TPP	PDH, alpha-KGDH, BCKD and transketolase	Defects link energy failure to neurological/cardiac disease; transketolase is in PPP.
B2, FAD/FMN	Flavoprotein redox reactions, respiratory chain	Not the cofactor for methylmalonyl-CoA mutase.
B3, NAD/NADP	Oxidation/reduction and reductive synthesis	Pellagra pattern: dermatitis, diarrhea and neuropsychiatric dysfunction.
B5, CoA / phosphopantetheine	Acyl transfer and fatty-acid synthase ACP domain	Connects acetyl-CoA, succinyl-CoA and fatty-acyl transport.
B6, PLP	Transamination, selected decarboxylation, ALA synthase, glycogen phosphorylase	Amino-acid, heme and glycogen questions can share this cofactor.
Biotin	Pyruvate, acetyl-CoA and propionyl-CoA carboxylases	Carries activated CO ₂ ; deficiency can affect several carboxylases.
Folate, THF	One-carbon transfer in purine and thymidylate synthesis	Deficiency can cause megaloblastic anemia.
B12	Methionine synthase and methylmalonyl-CoA mutase	Deficiency can raise homocysteine and methylmalonate and cause neurological dysfunction.
C, ascorbate	Collagen hydroxylation and iron handling	Scurvy reflects defective connective tissue.
A, retinal / retinoic acid	Vision / gene regulation and epithelial differentiation	Night blindness or epithelial changes; retinoic acid cannot restore retinal for vision.
D, calcitriol	Calcium/phosphate handling and transcription	Liver 25-hydroxylation then kidney 1-alpha-hydroxylation; deficiency impairs mineralization.
E, tocopherols	Lipid membrane antioxidant protection	Fat-malabsorption states can cause deficiency with neurological/hemolytic manifestations.
K	Gamma-carboxylation of clotting proteins and proteins C/S	Required for calcium-binding Gla residues; impaired vitamin-K recycling is a distinct mechanism.

B12 absorption requires release from food, transfer from binding proteins to intrinsic factor, and ileal uptake. Folate can improve the anemia of B12 deficiency without correcting its neurological injury, so the two deficiencies must be distinguished. Fat-soluble vitamins travel with dietary lipid and are vulnerable to cholestasis, pancreatic insufficiency and chylomicron transport defects. This review

intentionally omits supplement doses: the local PDF has OCR unit artifacts and is not a current prescribing source.

Do not confuse deficiency markers

Both folate and B12 deficiency can increase homocysteine. Increased methylmalonate supports B12-related dysfunction rather than isolated folate deficiency. The local book summary contains a B2/B12 text defect for methylmalonyl-CoA mutase; the detailed reaction at PDF p.361 confirms B12.

B12 transport stage	Required event	Failure point
Stomach	Food-bound B12 is released by gastric conditions and binds R-protein/haptocorrin	Reduced gastric acid impairs food-cobalamin release; do not equate all malabsorption with autoimmunity.
Duodenum	Pancreatic proteases degrade R-protein, allowing B12 to bind intrinsic factor	Pancreatic dysfunction can impair this transfer.
Ileum	Intrinsic-factor-B12 binds specific mucosal receptors and is taken up	Ileal disease/resection or loss of IF disrupts receptor-mediated uptake.
Blood and tissues	B12 travels on transcobalamin and reaches tissues; major stores are hepatic	Transport defects differ from dietary absence.
Enterohepatic recycling	Biliary B12 can be efficiently reclaimed in ileum	Large stores and recycling explain why dietary deficiency may take years.

Pernicious anemia specifically reflects autoimmune intrinsic-factor/parietal-cell-related failure, not every cause of B12 malabsorption. In methionine-synthase failure, folate becomes functionally trapped as methyl-THF, depleting forms required for purine/thymidylate synthesis. This links B12 deficiency to megaloblastosis while the second B12-dependent reaction explains increased methylmalonate.

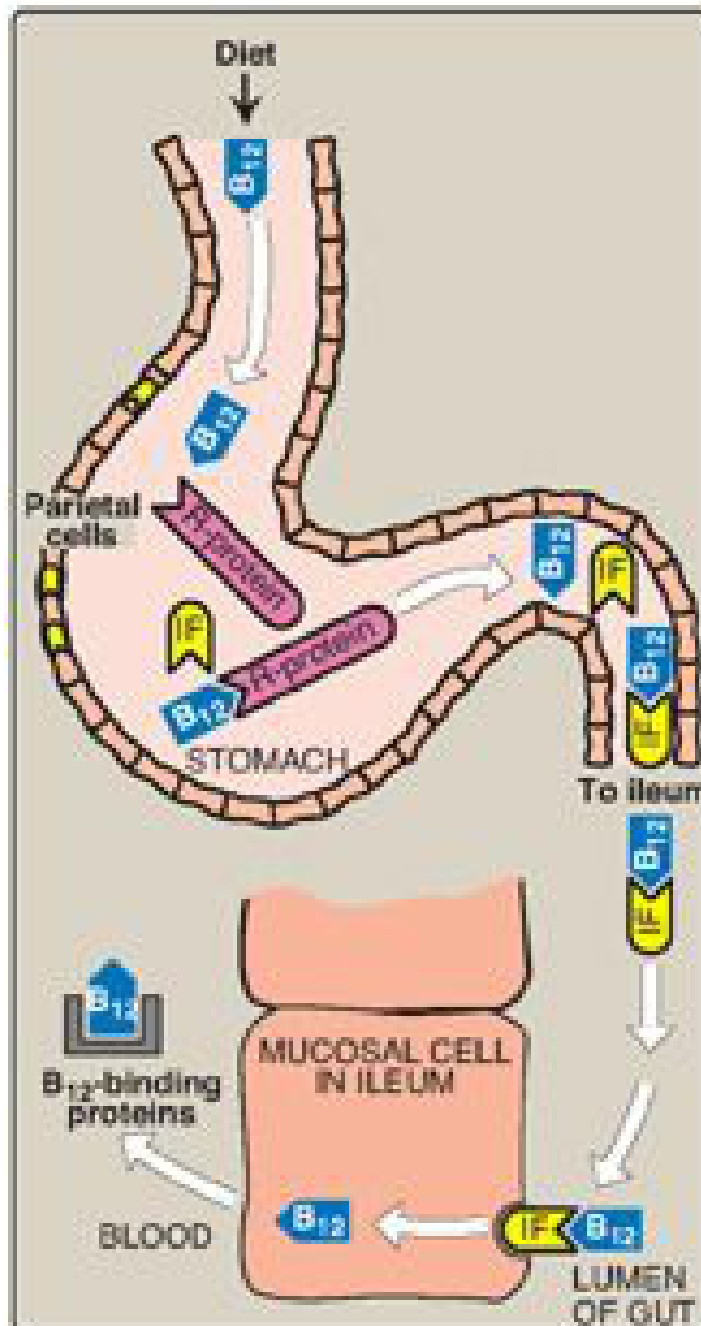


Figure 28.7, local PDF p.688 (caption begins p.687): follow food B₁₂, gastric binding proteins, pancreatic release, intrinsic-factor binding and ileal uptake.

Identify: Stomach; R-protein; Intrinsic factor; Pancreas; Ileum; Blood transport

Figure source: Lippincott Biochemistry 6e, PDF p. 688, extracted source diagram

Sources for this section

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - Ch.28 pp.681-720; Ch.20 pp.483-489

Nutrition, nitrogen balance and final integration checkpoints

SOURCE BASIS: book extension

Food energy is conventionally approximated as 4 kcal/g for carbohydrate or protein and 9 kcal/g for fat. Energy expenditure includes resting metabolism, activity and the thermic effect of food. These are population-level accounting values rather than a promise that every molecule yields an identical quantity of useful ATP. Protein quality depends on essential-amino-acid supply and digestibility. Combining complementary food sources can improve the overall amino-acid pattern.

Positive nitrogen balance accompanies net tissue growth or recovery. Negative balance occurs when losses exceed intake during inadequate protein supply or catabolic stress. Protein-energy malnutrition can combine energy and protein deficits in different proportions. Edema in a protein-deficient phenotype relates partly to reduced plasma protein and altered fluid distribution, but real malnutrition is more complex than a single albumin value.

Question to ask of any pathway	A complete answer should name
Where?	Tissue and subcellular compartment, including required membrane transport.
What goes in and out?	Carbon count, nitrogen source, energy and reducing-cofactor balance.
What controls flux?	Regulated enzyme, allosteric signals, hormonal modification and tissue exception.
What happens if blocked?	Accumulated substrate, missing product, compensatory pathway and clinical consequence.
What is the source confidence?	Lecture-supported versus book-only extension, plus any edition or transcript discrepancy.

Check yourself: A question gives fasting hypoglycemia. How do ketones refine the metabolic differential?

High ketones indicate that fat mobilization and hepatic ketogenesis are active, as in ordinary fasting or some carbohydrate-pathway disorders. Inappropriately low ketones suggest impaired fatty-acid oxidation/transport or inappropriate insulin effect. Interpret age, fasting duration, lactate, liver/muscle findings and other metabolites before assigning a specific disorder.

Check yourself: An amino acid contributes to both urea and glucose. Must its nitrogen and carbon follow the same route?

No. Transamination separates nitrogen handling from the fate of the carbon skeleton. Nitrogen can reach glutamate, ammonia or aspartate and then urea, while the alpha-ketoacid enters gluconeogenesis or oxidation. This separation is the organizing principle of amino-acid catabolism.

Sources for this section

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - Ch.27 pp.651-680

Coverage limits and next sources

- No original complete carbohydrate lecture deck found: Part 1 is seven handwritten pages and Part 2 is a six-page reconstructed summary, not slides 27-80 themselves.
- The three lipid recordings labelled 1.1, 1.2 and 1.3 together cover the first dietary-lipid lecture; later fatty-acid/lipoprotein sessions are referenced but not present.
- No Term 2 nitrogen, nucleotide, complex-lipid, vitamin or integrated-metabolism lecture files were found here. These sections are book extensions, not verified lecture scope.
- The local Lippincott edition differs from the lecturer reference. Page counters below refer to the inspected 6th-edition PDF.
- Transcript words and some handwritten claims conflict with biochemistry. Corrected teaching statements are identified rather than reproduced as true.

Lecture transcript archive

Full original files are embedded in this PDF. Open the PDF attachments panel in a reader that supports attachments, or use `_build/transcripts` in this folder. These are uncorrected source records, not an answer key.

Biochemistry Metabolism Recorded Lecture - 1h04m20s.transcript.txt

IMG_4406.transcript.txt

IMG_4407.transcript.txt

Lipid metabolism Esmaeili 1.1.transcript.txt

Lipid metabolism Esmaeili 1.2.transcript.txt

Lipid metabolism1 Esmaeili 1.3.transcript.txt

Local source inventory

Exact paths and source hashes are stored in `_build/source-manifest.json`. Section references distinguish original evidence from book additions.

Biochemistry [Lippincott's Illustrated Reviews Series] Denise R. Ferrier PhD.pdf

Biochemistry Metabolism (Part 1) Notes - Hawraa.pdf

Biochemistry metabolism notes (pt 2) slides 27-80 .pdf

Biochemistry Metabolism Recorded Lecture - 1h04m20s.transcript.txt

IMG_4406.transcript.txt

IMG_4407.transcript.txt

Lipid metabolism Esmaeili 1.1.transcript.txt

Lipid metabolism Esmaeili 1.2.transcript.txt

Lipid metabolism1 Esmaeili 1.3.transcript.txt